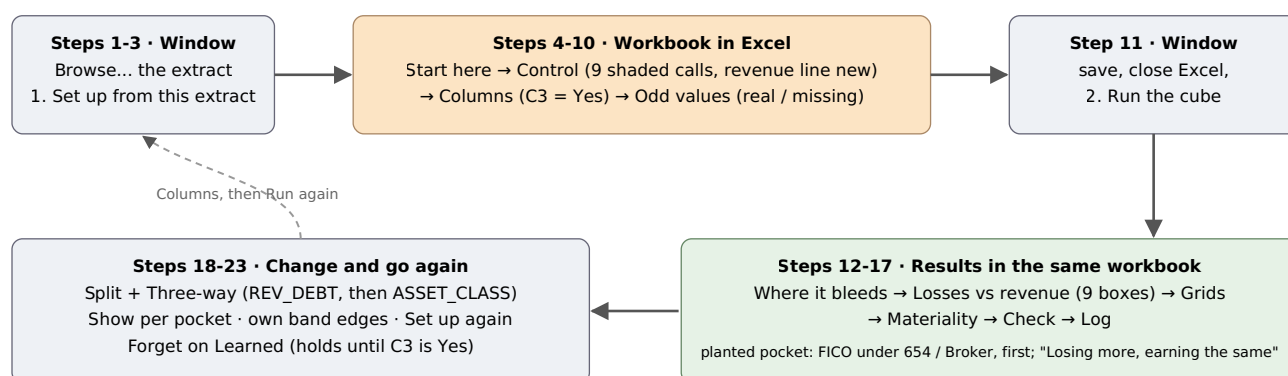


Origination analysis: the window and the workbook

Who this is for: the analyst preparing an origination analysis for a bank's line of business (LOB). No command line is needed. **What you finish with:** one workbook beside the loan extract. It holds your answers, a ranked list of the pockets that lose more than their share, losses set against revenue, and, if you ask for it, every pocket split by a third column and ranked again. **Walked:** 25 September 2026, on commit 1156ff4, with a synthetic book of 8,000 loans. The book hides two things: a bad pocket (scores under 620 in the Broker channel), and borrowers carrying more revolving debt than usual for their score, who go bad 1.8 times as often. RANR has nothing planted in it. No real bank data was used.

What changed since the third walk: Losses vs revenue has nine boxes set by lines on Control, a new Control call for revenue, and one chart per grid (steps 5, 9 and 13). The Split tab says what each grid holds fixed (step 18). The Three-way tab ranks every split pocket (steps 18 and 19). A Forget on Learned now holds until you confirm the column again (step 23).



Peach = where you answer (Excel). Grey = the window. Green = where you read the results. Nothing is typed at a command line.

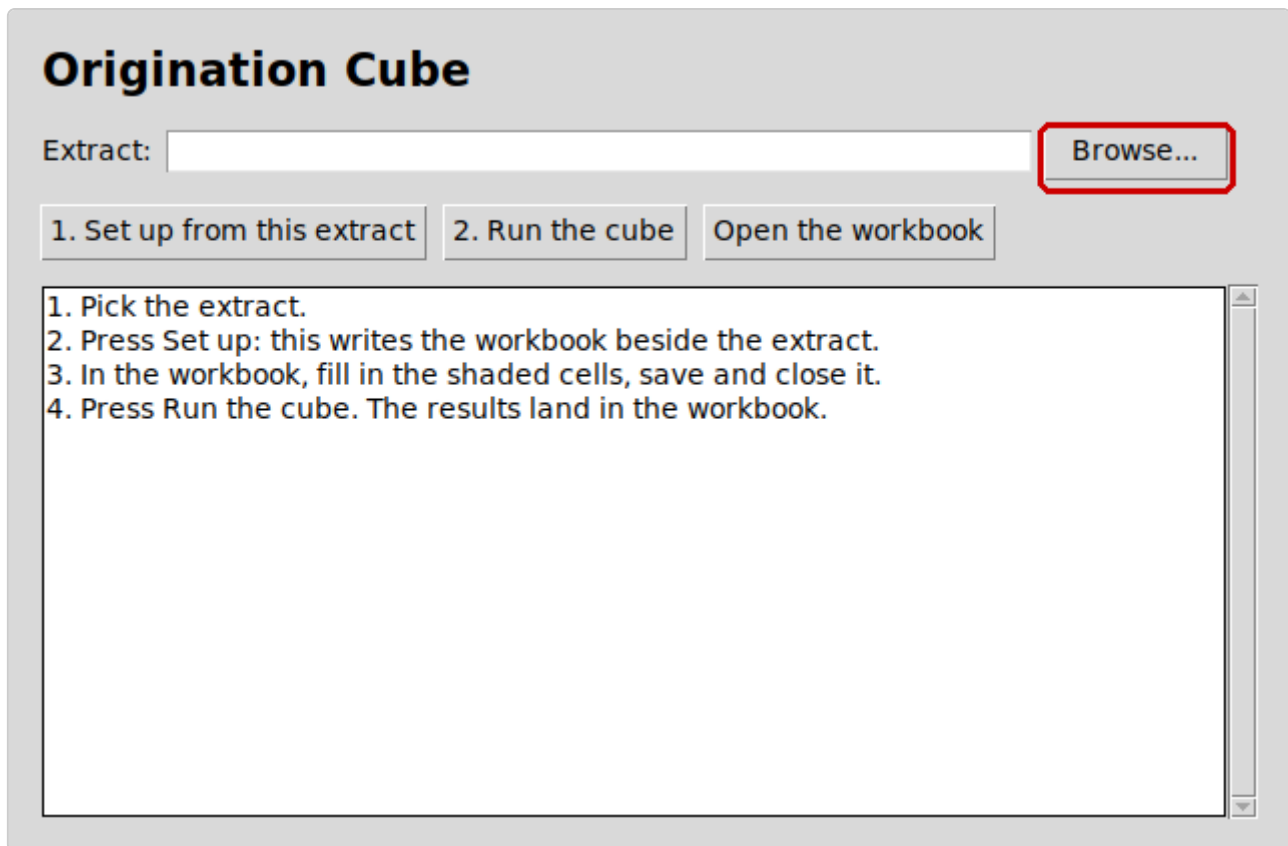
The route: two buttons in the window, one workbook in Excel.

Before you start

- **Python 3.10 or later**, and **Install add-ons.bat** run once. You only need to do this the first time.
- **Excel**.
- **The loan extract** (.csv or .xlsx) in its own folder. It needs a loan number, the booked amount, a yes/no outcome, GCO dollars and RANR dollars.

Step 1: Open the window

Double-click **Origination Cube.pyw** in the origination-cube folder.



Right: a window titled Origination Cube, an empty Extract box, three buttons, and four numbered lines in the box underneath.

Wrong: if the box says something needs installing, run `Install add-ons.bat` once, then open the window again.

Step 2: Pick the extract

Press **Browse...** and pick the loan file. After a first Set up the folder also holds the workbook. Pick the loan file, not the workbook; if you pick the workbook, Set up refuses and names the file to pick (wrong turn K).

Origination Cube

Extract: Browse...

1. Set up from Directory: 

1. Pick the ex

2. Press Set u

3. In the work

4. Press Run t

.claude

.claude-plugin

.git

.github

bls-laus-county-monitor

bureau-credit-risk-dashboard

canon

cashew-institute

cfpb-mortgage-mo

client-documents

consumer-credit-r

cowork-plugin

File name:

Files of type:

On Windows this is the usual Open dialog. The picture shows the same step on the test machine.

Origination Cube

Extract: Browse...

1. Set up from this extract 2. Run the cube Open the workbook

1. Pick the extract.

2. Press Set up: this writes the workbook beside the extract.

3. In the workbook, fill in the shaded cells, save and close it.

4. Press Run the cube. The results land in the workbook.

Right: the file name is at the right-hand end of the Extract box.

Step 3: Press 1. Set up from this extract

Origination Cube

Extract:

1. Set up from this extract

2. Run the cube

Open the workbook

Set up Consumer book Q3 - Origination Cube.xlsx from Consumer book Q3.csv:
8,000 loans, 9 columns.
7 columns to look at first, on the Columns tab.
Next: fill in the shaded cells on Control and Columns, save, close, and press Run.

Right: "Set up Consumer book Q3 - Origination Cube.xlsx from Consumer book Q3.csv: 8,000 loans, 9 columns", then "7 columns to look at first, on the Columns tab". The workbook now sits beside the extract.

Check the loan count against what the bank sent. If it's wrong, you picked the wrong file.

Step 4: Open the workbook: Start here

Press **Open the workbook**, or double-click it in its folder. It opens on **Start here**.

Origination Cube

Set up from Consumer book Q3.csv: 8,000 loans, 9 columns.

Step	Where	What to do
1	Control	Fill in the shaded cells: materiality, minimum loans, how much worse counts.
2	Columns	Check each column's meaning and whether to cut by it. Set C3 to Yes when done.
3	Odd values	Answer real or missing. Unanswered ones are used as is.
4	Launcher	Save and close this workbook, then press Run the cube.

Where things stand

Calls still to make on Control	9
Things to look at first on Columns	7
Odd values found	2
Last run	not run yet

Right: four steps (Control, Columns, Odd values, Launcher) and Where things stand: **9** calls to make on Control, 7 things to look at on Columns, 2 odd values, not run yet.

Step 5: Control: see what needs an answer

Click the **Control** tab.

Control center				
Fill in the shaded cells. The rest have starting values; change them if the population calls for it. Pick from the list, or type a number in the next column to override it. Changes apply on the next Run.				
Shaded = still needs an answer				
Setting	Choose	Or enter your own	In use	What it means
What the cube looks at				
How old a loan must be to count				Needs an answer before we run. Pick one, or enter your own.
How columns are recognised				
A number column with this many values or fewer is a category	12 values (recommended)		12 values	Term (36, 48, 60) or a grade from 1 to 10 is read as categories, not cut into bands.
A text column with more values than this is too fine to cut by	50 values (recommended)		50 values	State (50 or so) is still a dimension; a city or a dealer name is raised as a question instead.
How many bands a number column is cut into	5 bands (recommended)		5 bands	Fine enough to see a trend, with each band holding a fifth of the loans.
Where the band edges fall	Each band holds about the same number of loans (recommended)	n/a	Each band holds about the same number of loans	Every band can be tested equally well. Edges land on odd numbers like 663.
Is it enough loans to matter				
Fewest loans in a pocket before it is tested				Needs an answer before we run. Pick one, or enter your own.
Fewest loans with a loss before a loss rate is tested				Needs an answer before we run. Pick one, or enter your own.
Is it material				
Smallest excess loss worth reporting				Needs an answer before we run. Pick one, or enter your own.
Does it underperform				
What a pocket is judged against		n/a		Needs an answer before we run. Pick one from the list.
How much worse than its comparison a pocket must be				Needs an answer before we run. Pick one, or enter your own.
How much better than its comparison a pocket must be				Needs an answer before we run. Pick one, or enter your own.
How far revenue must move before it counts as more or less				Needs an answer before we run. Pick one, or enter your own.
Is it real				
How sure a difference must be before it counts				Needs an answer before we run. Pick one, or enter your own.
How often a real gap should be caught	80% of the time (recommended)		80% of the time	The usual standard. Used only to say how big a gap each pocket's size could show; it never flags anything.
Allowing for testing many pockets at once	Hold down the share of lucky finds (Benjamini-Hochberg) (recommended)	n/a	Hold down the share of lucky finds (Benjamini-Hochberg)	Of the pockets that pass, about 5% at 95% are expected to be luck. Suits scanning for leads.
How much better than its comparison a pocket must be				Needs an answer before we run. Pick one, or enter your own.
How far revenue must move before it counts as more or less				Needs an answer before we run. Pick one, or enter your own.
Is it real				

Right: nine rows are shaded peach. They are the calls the tool won't make for you: loan age, fewest loans, fewest losses, materiality, what a pocket is judged against, how much worse, how much better, **how far revenue must move** (ringed, new), and how sure. The other rows start on a recommended setting.

Step 6: Columns: see what each column was taken to be

Click the **Columns** tab.

Columns

Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.

Checked every column?

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (use ; between)	Show per pocket	Split pockets by it?	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	Loan number (key)						name contains 'loan'; a different value on every row	0%	L0000000, L0000001, L0000002
FICO	FICO score	Yes					name contains 'fico'; 98% of values between 300 and 850	0%	-9999, 713, 716
CHANNEL	Category	Yes					3 different values	0%	Online, Broker, Branch
ORIG_BAL BAD_FLAG	Booked amount Outcome (yes/no)	Yes					1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.	0%	7655.75, 27999.47, 39684.43
GCO_AMT	GCO dollars						1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.	0%	0.0, #N/A, 27620.62
RANR_AMT	RANR dollars						2,612 rows look odd (negatives). Used as it is until you answer it on the Odd values tab.	0%	-165.2, 54.71, 38.01
ASSET_CLASS	Category	Yes					suggested as Category from its shape alone (4 different values). If it's something more specific, say so and it will be remembered.	0%	1, 2, 4
REV_DEBT	Amount or number	Yes					suggested as Amount or number from its shape alone (numbers with 7,889 values). If it's something more specific, say so and it will be remembered.	0%	15868.86, 12884.7, 8920.97

Columns

Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.

Checked every column?

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (use ; between)	Show per pocket	Split pockets by it?	Look first
LOAN_NBR	Loan number (key)					
FICO	FICO score	Yes				160 rows look odd (repeated value). Used as it is until you answer it on the Odd values tab.
CHANNEL	Category	Yes				suggested as Category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.
ORIG_BAL BAD_FLAG	Booked amount Outcome (yes/no)	Yes				1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.
GCO_AMT	GCO dollars					1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.
RANR_AMT	RANR dollars					2,612 rows look odd (negatives). Used as it is until you answer it on the Odd values tab.
ASSET_CLASS	Category	Yes				suggested as Category from its shape alone (4 different values). If it's something more specific, say so and it will be remembered.
REV_DEBT	Amount or number	Yes				suggested as Amount or number from its shape alone (numbers with 7,889 values). If it's something more specific, say so and it will be remembered.

Right: one row per column, with a meaning in What it is and the reason beside it. Seven rows are shaded under **Look first**. Checked every column? (C3) is shaded and empty. **Split pockets by it?** (ringed) is for steps 18 and 19. Leave it blank for now.

Step 7: Odd values, and Learned

Click **Odd values**.

Odd values

Values that might be codes rather than real numbers. Answer real or missing (missing is treated as blank and counted). Unanswered ones are used as is.

Column	What looks odd	Rows	Answer	Note
FICO	the value -9999 repeated far more than any other	160		
RANR_AMT	negative values in a column that is mostly positive	2,612		

Right: FICO's -9999 (160 rows) and RANR's negatives (2,612 rows), each with a shaded Answer cell.

The **Learned** tab lists what the tool remembers from earlier runs. On the first extract it says Nothing learned yet.

Learned

Meanings and answers carried over from past runs. Set a row to Forget to drop it on the next Run. To stop it being learned again, also change it on Columns.

Keep?	Kind	Column	What it learned	First confirmed	Last confirmed	Times
			Nothing learned yet.			

Step 8: If you press Run too early

If you save, close and press **2. Run the cube** before answering, nothing runs.

Origination Cube

Extract: -4db589c3cc31/scratchpad/w4/walk/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Control!C6: "How old a loan must be to count" needs an answer. Pick one, or enter your own in column D.
- Control!C13: "Fewest loans in a pocket before it is tested" needs an answer. Pick one, or enter your own in column D.
- Control!C14: "Fewest loans with a loss before a loss rate is tested" needs an answer. Pick one, or enter your own in column D.
- Control!C16: "Smallest excess loss worth reporting" needs an answer. Pick one, or enter your own in column D.
- Control!C18: "What a pocket is judged against" needs an answer. Pick one from the list.
- Control!C19: "How much worse than its comparison a pocket must be" needs an answer. Pick one, or enter your own in column D.
- Control!C20: "How much better than its comparison a pocket must be" needs an answer. Pick one, or enter your own in column D.
- Control!C21: "How far revenue must move before it counts as more or less"

Right: "Couldn't run yet", then one line per thing to fix, each naming the tab and cell. The same list goes to the **Log** tab.

Scroll the box. The last three lines (Control!C21, Control!C23 and Columns!C3) are below the fold:

Origination Cube

Extract: -4db589c3cc31/scratchpad/w4/walk/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Control!C17: "Fewest loans with a loss before a loss rate is tested" needs an answer. Pick one, or enter your own in column D.

- Control!C16: "Smallest excess loss worth reporting" needs an answer. Pick one, or enter your own in column D.
- Control!C18: "What a pocket is judged against" needs an answer. Pick one from the list.
- Control!C19: "How much worse than its comparison a pocket must be" needs an answer. Pick one, or enter your own in column D.
- Control!C20: "How much better than its comparison a pocket must be" needs an answer. Pick one, or enter your own in column D.
- Control!C21: "How far revenue must move before it counts as more or less" needs an answer. Pick one, or enter your own in column D.
- Control!C23: "How sure a difference must be before it counts" needs an answer. Pick one, or enter your own in column D.
- Columns!C3: set Checked every column to Yes once you've checked each column's meaning.

Step 9: Fill in the shaded cells on Control

Click each shaded **Choose** cell and pick from its list. To use a number that isn't offered, type it in **Or enter your own** instead. The walk used:

Setting	Answer
How old a loan must be to count	Every loan
Fewest loans in a pocket before it is tested	30 loans
Fewest loans with a loss before a loss rate is tested	10 losses
Smallest excess loss worth reporting	1% of the book's total losses
What a pocket is judged against	The rest of its band
How much worse	1.25 times
How much better	0.8 times
How far revenue must move before it counts	What luck alone can move it (suggested)
How sure	95% sure

Control center				
Fill in the shaded cells. The rest have starting values; change them if the population calls for it. Pick from the list, or type a number in the next column to override it. Changes apply on the next Run.				
Shaded = still needs an answer				
Setting	Choose	Or enter your own	In use	What it means
What the cube looks at				
How old a loan must be to count	Every loan		Every loan	New loans have not had time to go bad, so a pocket full of recent loans looks better than it is.
How columns are recognised				
A number column with this many values or fewer is a category	12 values (recommended)		12 values	Term (36, 48, 60) or a grade from 1 to 10 is read as categories, not cut into bands.
A text column with more values than this is too fine to cut by	50 values (recommended)		50 values	State (50 or so) is still a dimension; a city or a dealer name is raised as a question instead.
How many bands a number column is cut into	5 bands (recommended)		5 bands	Fine enough to see a trend, with each band holding a fifth of the loans.
Where the band edges fall	Each band holds about the same number of loans (recommended)	n/a	Each band holds about the same number of loans	Every band can be tested equally well. Edges land on odd numbers like 663.
Is it enough loans to matter				
Fewest loans in a pocket before it is tested	30 loans		30 loans	The old workbook's value. Stops a test on a handful of loans; the test itself decides whether a gap is luck.
Fewest loans with a loss before a loss rate is tested	10 losses		10 losses	A loss rate rests on how many loans lost money, not how many a pocket holds. Ten is a common floor.
Is it material				
Smallest excess loss worth reporting	1% of the book's total losses		1% of the book's total losses	Scales with the book, so the same setting works on any population.
Does it underperform				
What a pocket is judged against	The rest of its band	n/a	The rest of its band	Broker under 620 against the rest of under 620. Both comparisons are printed; this one decides the flag.
How much worse than its comparison a pocket must be	1.25 times		1.25 times	The old workbook's value. Moderate gaps; more pockets flagged.
How much better than its comparison a pocket must be	0.8 times		0.8 times	The old workbook's value.
How far revenue must move before it counts as more or less	What luck alone can move it (suggested)		What luck alone can move it (suggested)	Worked out from this book: the smallest revenue gap a typical pocket can tell apart from luck. Inside it, revenue reads as about the same.
Is it real				
How sure a difference must be before it counts	95% sure		95% sure	The usual standard. About 1 pocket in 20 with no real difference will still pass.
How often a real gap should be caught	80% of the time (recommended)		80% of the time	The usual standard. Used only to say how big a gap each pocket's size could show; it never flags anything.
Allowing for testing many pockets at once	Hold down the share of lucky finds (Benjamini-Hochberg) (recommended)	n/a	Hold down the share of lucky finds (Benjamini-Hochberg)	Of the pockets that pass, about 5% at 95% are expected to be luck. Suits scanning for leads.
How much better than its comparison a pocket must be	0.8 times		0.8 times	The old workbook's value.
How far revenue must move before it counts as more or less	What luck alone can move it (suggested)		What luck alone can move it (suggested)	Worked out from this book: the smallest revenue gap a typical pocket can tell apart from luck. Inside it, revenue reads as about the same.
Is it real				

Right: the peach shading is gone, and In use and What it means fill in for each row.

The revenue line has no number yet. With What luck alone can move it, the line is worked out when you Run. It shows at the top of Losses vs revenue and on Check (1.25x on this book). The other options:

- 5 percent either way or 10 percent either way: revenue counts as more or less once it's 5% or 10% off.
- The same lines as for losses: revenue uses the two answers above it, turned round (0.80x and 1.25x here).
- Or enter your own: a share between 0.01 and 0.9, such as 0.15 for 15%.

Step 10: Confirm the columns and answer the odd values

On **Columns**, check each What it is. Fix any that's wrong from its list, then set **Checked every column?** (C3) to **Yes**.

Columns									
Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.									
Checked every column? Yes									
Column	What it is	Cut by it? Yes means (outcome only)	Band edges (use ; between)	Show per pocket	Split pockets by it?	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	Loan number (key)						name contains 'loan'; a different value on every row	0%	L0000000, L0000001, L0000002
FICO	FICO score	Yes				160 rows look odd (repeated value). Used as it is until you answer it on the Odd values tab.	name contains 'fico'; 98% of values between 300 and 850	0%	-9999, 713, 716
CHANNEL	Category	Yes				suggested as Category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.	3 different values	0%	Online, Broker, Branch
ORIG_BAL	Booked amount	Yes					name contains 'bal'; positive on every loan	0%	7655.75, 27999.47, 39684.43
BAD_FLAG	Outcome (yes/no)					1 value is neither 0 nor 1. They're left out of the outcome rates and counted, if they mean something, fix the extract.	name contains 'bad'; 0 and 1 (7.1% are 1)	0%	0, 2, 1
GCO_AMT	GCO dollars					1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.	name contains 'gco'; zero on 93% of loans, never negative	0%	0.0, #N/A, 27620.62
RANR_AMT	RANR dollars					2,612 rows look odd (negatives). Used as it is until you answer it on the Odd values tab.	name contains 'ranr'; numbers	0%	-165.2, 54.71, 38.01
ASSET_CLASS	Category	Yes				suggested as Category from its shape alone (4 different values). If it's something more specific, say so and it will be remembered.	4 different values	0%	1, 2, 4
REV_DEBT	Amount or number	Yes				suggested as Amount or number from its shape alone (numbers with 7,889 values). If it's something more specific, say so and it will be remembered.	numbers with 7,889 values	0%	15868.86, 12884.7, 8920.97

On **Odd values**, answer each row. The walk said FICO's -9999 is **missing** and RANR's negatives are **real**, because RANR is revenue and can be negative.

Odd values				
Values that might be codes rather than real numbers. Answer real or missing (missing is treated as blank and counted). Unanswered ones are used as is.				
Column	What looks odd	Rows	Answer	Note
FICO	the value -9999 repeated far more than any other negative values in a column that is mostly positive		160 missing	
RANR_AMT			2,612 real	

Step 11: Save, close, and press 2. Run the cube

Save the workbook and **close Excel**. Then press **2. Run the cube**.

Origination Cube

Extract:

Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.
Worst for RANR per booked dollar: REV_DEBT 6,559 to under 9,837 / CHANNEL Broker.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds

Right:

- "Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree."
- one Worst for line per measure. For the three loss measures it's **FICO under 654 / CHANNEL Broker**, the planted pocket seen through five equal bands.

The Worst for RANR line (REV_DEBT 6,559 to under 9,837 / Broker) is noise. The test data plants nothing in RANR.

Step 12: Read Where it bleeds

Open the workbook again. **Where it bleeds** lists every pocket that loses more than its share, largest first.

Where it bleeds

Pockets losing more than their share (RANR: earning less), target first. The flag compares each pocket with the rest of its band. Red: worse. Amber: worse, but could be luck. "Luck alone" is how often a gap this big turns up with no real difference behind it.

Measure	Band column	Band	Segment column	Segment	Loans	Rate	Book rate	Excess	Excess is in	Material	Vs rest of book	Luck alone	Vs rest of band	Luck alone	Flag (vs the rest of its band)	Smallest gap it could show
Outcome, share of loans	FICO	under 654	CHANNEL	Broker	523	24.47%	7.13%	90.7 loans	yes		4.14x under 0.01%	2.38x under 0.01%			1.47x or more	
Outcome, share of loans	REV DEBT	17,187 and over	CHANNEL	Broker	507	22.29%	7.13%	76.9 loans	yes		3.65x under 0.01%	2.01x under 0.01%			1.48x or more	
Outcome, share of loans	FICO	under 654	ASSET CLASS	4	392	20.41%	7.13%	52.1 loans	yes		3.17x under 0.01%	1.54x 3.8% worse			1.55x or more	
Outcome, share of loans	REV DEBT	17,187 and over	ASSET CLASS	4	424	18.36%	7.13%	48.8 loans	yes		2.87x under 0.01%	1.41x 8.2% worse, but could be luck			1.53x or more	
Outcome, share of loans	REV DEBT	17,187 and over	ASSET CLASS	1	411	14.36%	7.13%	29.7 loans	yes		2.13x under 0.01%	0.98x 95.1% in line			1.53x or more	
Outcome, share of loans	FICO	under 654	ASSET CLASS	3	350	15.14%	7.13%	28.1 loans	yes		2.24x under 0.01%	1.01x 98.1% in line			1.58x or more	
Outcome, share of loans	FICO	under 654	ASSET CLASS	1	420	13.57%	7.13%	27.1 loans	yes		2.03x 0.0%	0.87x 67.5% in line			1.53x or more	
Outcome, share of loans	REV DEBT	17,187 and over	ASSET CLASS	3	380	14.21%	7.13%	26.8 loans	yes		2.10x 0.0%	0.96x 93.2% in line			1.56x or more	
Outcome, share of loans	ORIG_BAL	15,449 to under 26,324	CHANNEL	Broker	537	17.73%	7.13%	24.7 loans	yes		1.73x 0.6%	1.58x 5.6% worse, but could be luck			1.46x or more	
Outcome, share of loans	REV DEBT	17,187 and over	CHANNEL	Branch	525	11.62%	7.13%	23.6 loans	yes		1.71x 0.1%	0.72x 6.3% better, but could be luck			1.47x or more	
Outcome, share of loans	REV DEBT	17,187 and over	CHANNEL	Online	568	10.56%	7.13%	19.5 loans	yes		1.54x 0.7%	0.63x 0.2% better			1.45x or more	
Outcome, share of loans	ORIG_BAL	37,951 to under 49,152	ASSET CLASS	4	414	11.59%	7.13%	18.5 loans	yes		1.68x 2.4%	1.79x 6.4% worse, but could be luck			1.53x or more	
Outcome, share of loans	FICO	under 654	CHANNEL	Online	531	10.36%	7.13%	17.2 loans	yes		1.50x 1.8%	0.59x 0.0% better			1.47x or more	
Outcome, share of loans	FICO	654 to under 686	ASSET CLASS	4	410	11.22%	7.13%	16.8 loans	yes		1.62x 1.2%	1.56x 12.5% worse, but could be luck			1.54x or more	
Outcome, share of loans	FICO	under 654	ASSET CLASS	2	393	11.20%	7.13%	16.0 loans	yes		1.62x 1.5%	0.68x 9.0% better, but could be luck			1.55x or more	
Outcome, share of loans	REV DEBT	13,051 to under 17,187	ASSET CLASS	4	411	10.95%	7.13%	15.7 loans	yes		1.58x 1.9%	1.51x 15.2% worse, but could be luck			1.53x or more	
Outcome, share of loans	REV DEBT	13,051 to under 17,187	CHANNEL	Broker	553	9.95%	7.13%	15.6 loans	yes		1.44x 2.8%	1.37x 22.3% worse, but could be luck			1.46x or more	
Outcome, share of loans	FICO	under 654	CHANNEL	Branch	501	10.18%	7.13%	15.3 loans	yes		1.47x 2.9%	0.59x 0.0% better			1.48x or more	
Outcome, share of loans	REV DEBT	17,187 and over	ASSET CLASS	2	385	10.91%	7.13%	14.6 loans	yes		1.57x 2.3%	0.69x 8.2% better, but could be luck			1.55x or more	
Outcome, share of loans	ORIG_BAL	15,449 to under 26,324	ASSET CLASS	4	413	10.41%	7.13%	13.6 loans	yes		1.50x 8.6%	1.25x 39.8% in line			1.53x or more	
Outcome, share of loans	ORIG_BAL	37,951 to under 49,152	CHANNEL	Broker	531	9.23%	7.13%	11.2 loans	yes		1.32x 20.8%	1.30x 25.5% worse, but could be luck			1.47x or more	
Outcome, share of loans	ORIG_BAL	26,324 to under 37,951	ASSET CLASS	4	412	9.71%	7.13%	10.6 loans	yes		1.39x 19.6%	1.80x 7.1% worse, but could be luck			1.53x or more	
Outcome, share of loans	ORIG_BAL	15,449 to under 26,324	ASSET CLASS	3	427	9.37%	7.13%	9.6 loans	yes		1.34x 24.3%	1.08x 90.8% in line			1.52x or more	
Outcome, share of loans	FICO	654 to under 686	CHANNEL	Branch	503	8.95%	7.13%	9.2 loans	yes		1.28x 16.9%	1.13x 82.1% in line			1.48x or more	
Outcome, share of loans	ORIG_BAL	15,449 to under 26,324	ASSET CLASS	1	404	9.16%	7.13%	8.2 loans	yes		1.31x 27.9%	1.04x 97.7% in line			1.54x or more	
Outcome, share of loans	ORIG_BAL	49,152 and over	CHANNEL	Broker	520	8.46%	7.13%	6.9 loans	yes		1.20x 44.1%	1.90x 5.2% worse, but could be luck			1.47x or more	
Outcome, share of loans	ORIG_BAL	37,951 to under 49,152	ASSET CLASS	3	382	8.90%	7.13%	6.8 loans	yes		1.26x 35.9%	1.19x 59.1% in line			1.56x or more	
Outcome, share of loans	ORIG_BAL	under 15,449	CHANNEL	Broker	514	8.37%	7.13%	6.4 loans	yes		1.19x 44.8%	1.42x 20.1% worse, but could be luck			1.47x or more	
Outcome, share of loans	FICO	654 to under 686	CHANNEL	Broker	531	8.10%	7.13%	5.2 loans	below the line		1.15x 42.6%	0.97x 87.3% in line			1.47x or more	
Outcome, share of loans	ORIG_BAL	under 15,449	ASSET CLASS	4	425	8.24%	7.13%	4.7 loans	below the line		1.17x 51.6%	1.34x 35.8% worse, but could be luck			1.53x or more	
Outcome, share of loans	ORIG_BAL	37,951 to under 49,152	CHANNEL	Branch	526	7.98%	7.13%	4.5 loans	below the line		1.13x 57.6%	1.03x 88.8% in line			1.47x or more	
Outcome, share of loans	FICO	654 to under 686	ASSET CLASS	3	377	8.22%	7.13%	4.1 loans	below the line		1.16x 51.8%	1.00x 98.1% in line			1.56x or more	

Where it bleeds

Pockets losing more than their share (RANR: earning less), target first. The flag compares each pocket with the rest of its band. Red: worse. Amber: worse, but could be luck. "Luck alone" is how often a gap this big turns up with no real difference behind it.

Measure	Band column	Band	Segment column	Segment	Loans	Rate	Book rate	Excess	Excess is in	Material	Vs rest of book	Luck alone	Vs rest of band	Luck alone	Flag (vs the rest of its band)	Smallest gap it could show
Outcome, share of loans	FICO	under 654	CHANNEL	Broker	523	24.47%	7.13%	90.7 loans	yes		4.14x under 0.01%	2.38x under 0.01%			1.47x or more	
Outcome, share of loans	REV DEBT	17,187 and over	CHANNEL	Broker	507	22.29%	7.13%	76.9 loans	yes		3.65x under 0.01%	2.01x under 0.01%			1.48x or more	
Outcome, share of loans	FICO	under 654	ASSET CLASS	4	392	20.41%	7.13%	52.1 loans	yes		3.17x under 0.01%	1.54x 3.8% worse			1.55x or more	

Right: the first row is FICO under 654 / CHANNEL Broker: 523 loans, 24.47% bad against a book of 7.13%, an excess of 90.7 loans, 4.14x the rest of the book, 2.38x the rest of its band, flagged **worse**.

How to read a row:

- **Red rows** are worse, and the test says it isn't luck. **Amber** rows are worse but could be luck.
- **Excess is in** gives the unit: loans for the bad-loan share, dollars of a named column otherwise.
- **The flag** uses the comparison you chose in step 9 (here, the rest of its band). The heading says which.
- **Luck alone** is how often a gap this big turns up with no real difference. A figure shown as 0.0% is under 0.05%, not zero.
- **RANR** rows are shortfalls: pockets earning less than their share.

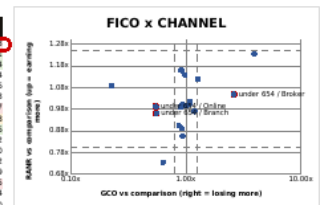
Step 13: Losses vs revenue

Losses vs revenue puts every pocket of at least 30 loans in one of nine boxes. Each side reads more, the same or less by the lines at the top of the tab, and the box is the pair. There's one block and one chart per grid.

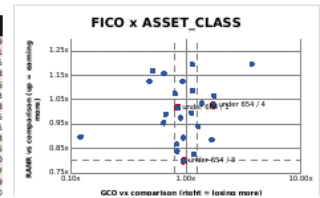
Losses vs revenue

Each pocket's GCO and RANR against the rest of its band. GCO counts as more at 1.25x or above and less at 0.80x or below. Revenue counts as more at 1.25x or above and less at 0.80x or below. what luck alone can move it: in a pocket of typical size, a revenue gap smaller than 1.25x can't be told from luck at 95% sure and 80% caught. Both are set on Control. Nothing is needed.

FICO x CHANNEL		Segment	Loans	GCO vs comparison	GCO flag	RANR vs comparison	RANR flag	Which box	GCO excess over the book	RANR short of the book (\$)
Losing more, earning the same-3: Losing less, earning less-1: About the same on both-2: Losing less, earning the same-3:										
under 654	Broker	523	2.62x worse	1.05x in line	1.05x in line	Losing more, earning the same	1,757,039	-2,623		
under 654	Online	531	0.55x better	0.99x in line	0.99x in line	Losing less, earning the same	271,012	-611		
under 654	Branch	502	0.55x better	0.96x in line	0.96x in line	Losing less, earning the same	249,020	584		
654 to under 686	Branch	503	1.19x in line	0.97x in line	0.97x in line	About the same on both	185,594	2,454		
654 to under 686	Broker	531	0.92x in line	0.89x in line	0.89x in line	About the same on both	63,685	5,466		
654 to under 686	Online	505	0.91x in line	1.16x in line	1.16x in line	About the same on both	53,816	-3,713		
(marked missing)	Broker	54	3.93x too few losses to test	1.24x in line	1.24x in line	Losing more, earning the same	22,556	-97		
(marked missing)	Online	55	0.63x too few losses to test	0.73x worse, but could be luck	0.73x worse, but could be luck	Losing less, earning less	-37,115	1,688		
(marked missing)	Branch	51	0.23x too few losses to test	1.09x in line	1.09x in line	Losing less, earning the same	-54,555	366		
686 to under 712	Broker	501	1.07x in line	1.00x in line	1.00x in line	About the same on both	-201,130	2,652		
686 to under 712	Online	546	1.00x in line	1.00x in line	1.00x in line	About the same on both	-236,102	3,050		
686 to under 712	Branch	515	0.94x in line	1.00x in line	1.00x in line	About the same on both	-242,718	2,662		
712 to under 746	Branch	536	1.09x in line	1.01x in line	1.01x in line	About the same on both	-256,765	-339		
746 and over	Branch	518	1.26x worse, but could be luck	1.12x in line	1.12x in line	Losing more, earning the same	-262,995	-8,405		
712 to under 746	Broker	509	0.94x in line	0.85x in line	0.85x in line	About the same on both	-289,277	5,594		
712 to under 746	Online	564	0.98x in line	1.14x in line	1.14x in line	About the same on both	-315,198	-4,770		
746 and over	Broker	536	0.90x in line	0.99x in line	0.99x in line	About the same on both	-352,303	-3,782		
746 and over	Online	517	0.87x in line	0.90x in line	0.90x in line	About the same on both	-353,483	-211		



FICO x ASSET CLASS									
Asset Class	Segment	Loans	GCO vs comparison	GCO flag	RANR vs comparison	RANR flag	Which box	GCO excess over the book	RANR short of the book (\$)
You are given the following data for each asset class. E: Losing the same, earning less; L: About the same on both; W: About the same on both; W: About the same on both; W: About the same on both.									
under 654	4	392	1.74x worse	1.03x in line	1.03x in line	Losing more, earning the same	1,039,185	-1,449	
under 654	3	350	0.95x in line	0.80x worse, but could be luck	0.80x worse, but could be luck	Losing the same, earning less	490,554	5,583	
under 654	1	420	0.85x in line	1.02x in line	1.02x in line	About the same on both	471,014	-1,275	
654 to under 686	4	410	1.68x worse, but could be luck	0.89x in line	0.89x in line	Losing more, earning the same	336,769	4,768	
under 654	2	394	0.65x better, but could be luck	1.16x in line	1.16x in line	Losing less, earning the same	276,317	-5,506	
654 to under 686	3	377	1.13x in line	1.19x in line	1.19x in line	About the same on both	125,432	-4,243	
(marked missing)	2	36	3.75x too few losses to test	1.19x in line	1.19x in line	Losing more, earning the same	24,504	-58	
(marked missing)	3	43	0.95x too few losses to test	0.80x in line	0.80x in line	About the same on both	-20,613	876	
(marked missing)	4	39	0.80x too few losses to test	1.07x in line	1.07x in line	About the same on both	-20,789	255	
(marked missing)	1	42	0.12x too few losses to test	0.90x in line	0.90x in line	Losing less, earning the same	-52,215	883	
654 to under 686	2	355	0.67x better, but could be luck	0.90x in line	0.90x in line	Losing less, earning the same	-71,183	1,235	
712 to under 746	4	427	1.76x worse, but could be luck	1.07x in line	1.07x in line	Losing more, earning the same	-86,398	-1,880	
654 to under 686	1	397	0.64x better, but could be luck	0.95x in line	0.95x in line	Losing less, earning the same	-87,924	2,447	
686 to under 712	1	372	1.28x worse, but could be luck	0.94x in line	0.94x in line	Losing more, earning the same	-97,370	5,589	
686 to under 712	4	407	1.18x in line	0.83x in line	0.83x in line	About the same on both	-137,394	7,730	



Losses vs revenue

Each pocket's GCO and RANR against the rest of its band. GCO counts as more at 1.25x or above and less at 0.80x or below. Revenue counts as more at 1.25x or above and less at 0.80x or below. what luck alone can move it: in a pocket of typical size, a revenue gap smaller than 1.25x can't be told from luck at 95% sure and 80% caught. Both are set on Control. Nothing is needed.

FICO x CHANNEL		Losing more, earning the same: 3; Losing less, earning less: 1; About the same on both: 11; Losing less, earning the same: 3							
Band	Segment	Loans	GCO vs comparison	GCO flag	RANR vs comparison	RANR flag	Which box	GCO excess over the book	RANR short of the book (\$)
under 654	Broker	523	2.62x worse		1.05x in line		Losing more, earning the same	1,757,039	-2,623
under 654	Online	531	0.55x better		0.99x in line		Losing less, earning the same	271,012	-611
under 654	Branch	502	0.55x better		0.96x in line		Losing less, earning the same	249,020	584
654 to under 686	Branch	503	1.19x in line		0.97x in line		About the same on both	185,594	2,454
654 to under 686	Broker	531	0.92x in line		0.89x in line		About the same on both	63,685	5,466
654 to under 686	Online	505	0.91x in line		1.16x in line		About the same on both	53,816	-3,713
(marked missing)	Broker	54	3.93x too few losses to test		1.24x in line		Losing more, earning the same	22,556	-97
(marked missing)	Online	55	0.63x too few losses to test		0.73x worse, but could be luck		Losing less, earning less	-37,115	1,688
(marked missing)	Branch	51	0.23x too few losses to test		1.09x in line		Losing less, earning the same	-54,555	366
686 to under 712	Broker	501	1.07x in line		1.00x in line		About the same on both	-201,130	2,652
686 to under 712	Online	546	1.00x in line		1.00x in line		About the same on both	-236,102	3,050
686 to under 712	Branch	515	0.94x in line		1.00x in line		About the same on both	-242,718	2,662
712 to under 746	Branch	536	1.09x in line		1.01x in line		About the same on both	-256,765	-339
746 and over	Branch	518	1.26x worse, but could be luck		1.12x in line		Losing more, earning the same	-262,995	-8,405
712 to under 746	Broker	509	0.94x in line		0.85x in line		About the same on both	-289,277	5,594
712 to under 746	Online	564	0.98x in line		1.14x in line		About the same on both	-315,198	-4,770
746 and over	Broker	536	0.90x in line		0.99x in line		About the same on both	-352,303	-3,782
746 and over	Online	517	0.87x in line		0.90x in line		About the same on both	-353,483	-211

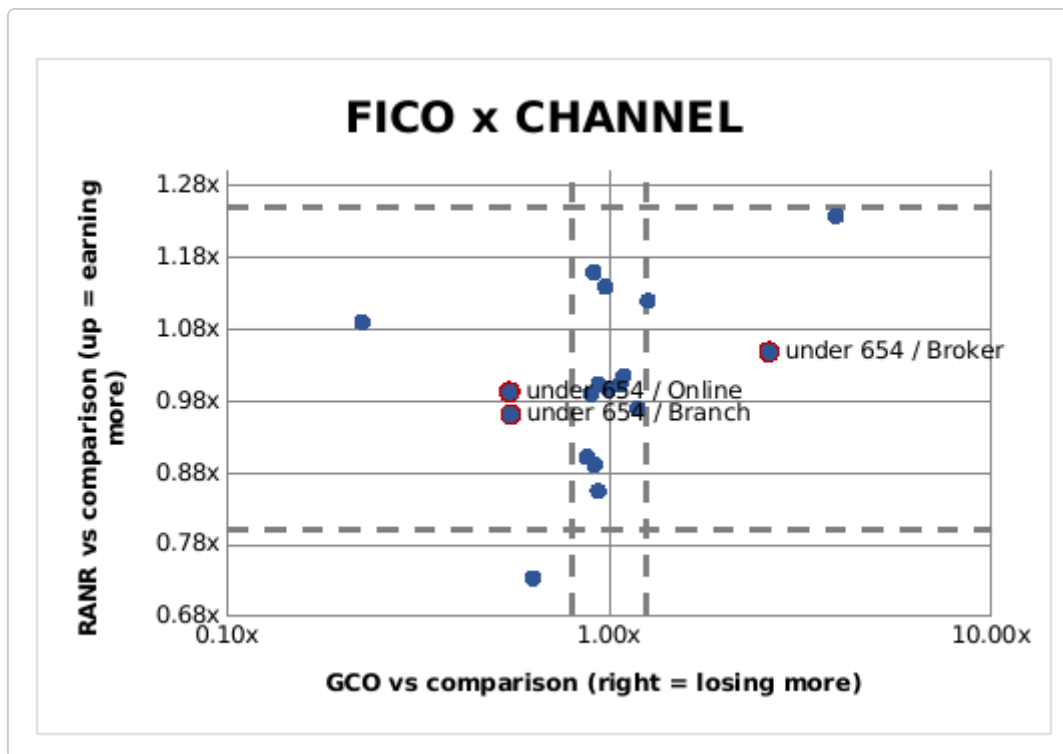
Right:

- The subtitle gives both sets of lines: GCO at 1.25x and 0.80x, revenue at 1.25x and 0.80x, "what luck alone can move it".
- The planted pocket, FICO under 654 / Broker (ringed), reads **Losing more, earning the same**: GCO 2.62x the rest of its band, RANR 1.05x.
- Above each block, a count per box.

Read the box with the dollars in mind. The box compares a pocket with the rest of its band. The two dollar columns and the order of the rows compare it with the book. So under 654 / Online reads **Losing less** (0.55x its band) but loses \$271,012 more than the book would expect. And 746 and over / Branch reads **Losing more** but loses \$262,995 less (defect 2).

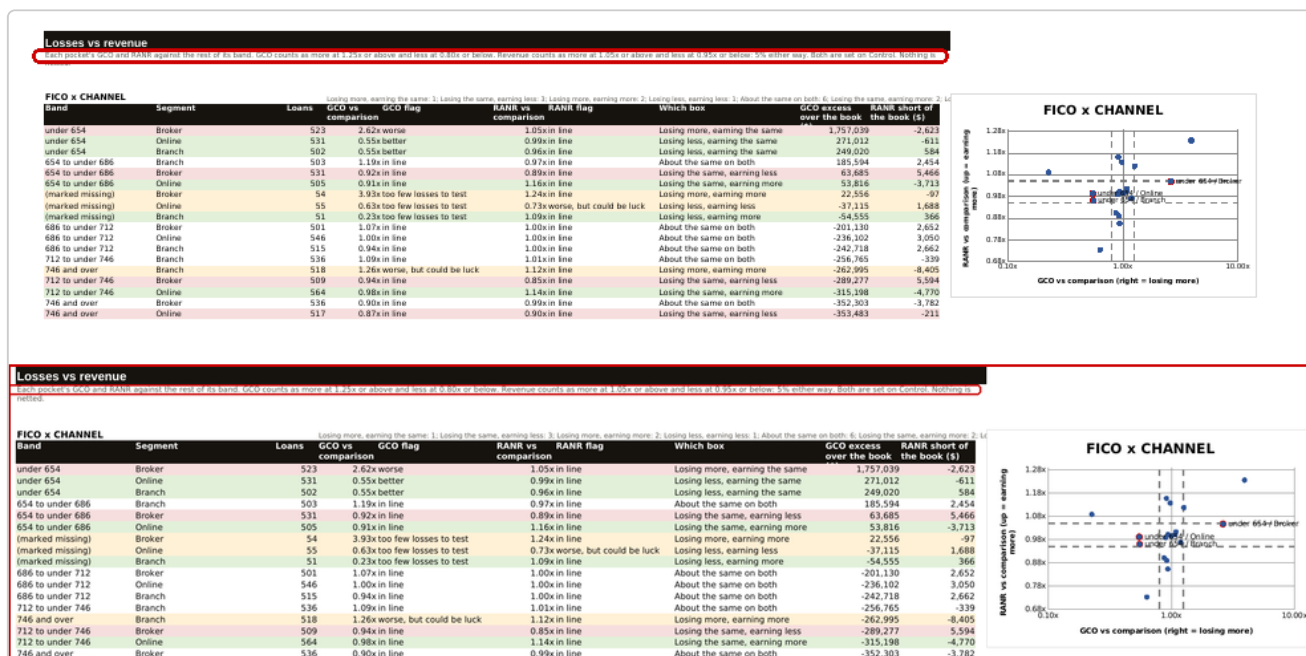
Pockets marked too few losses to test still get a box and a colour. Treat their box as unread (defect 3).

The chart plots the same pockets, GCO across on a scale of tens and RANR up, with the four Control lines dashed. The three pockets with the most GCO excess over the book are named:



Two of the three names on this chart sit left of the 0.80x line: they are big in dollars and better than the rest of their band. Where names overlap, use the table.

Try another revenue line. Change How far revenue must move on Control, save, close and Run. With 5 percent either way, 69 of 112 pockets read as earning more or less, against 5 with the luck line. The planted pocket stays in earning the same at 1.05x, just under the line:



What each option did on this book (112 pockets):

Revenue line	Revenue counts as more / less at	Earning more	Earning less
What luck alone can move it	1.25x / 0.80x	0	5
5 percent either way	1.05x / 0.95x	37	32

Revenue line	Revenue counts as more / less at	Earning more	Earning less
10 percent either way	1.10x / 0.90x	22	18
The same lines as for losses	1.25x / 0.80x	0	5
Your own: 0.15	1.15x / 0.85x	8	8

On this book luck and the same lines as for losses land on the same lines (1.248 against 1.25). The luck line depends on the grids in the run, so it moves when you split or change bands: 1.22x with ASSET_CLASS split, 1.24x with the LOB's own edges (defect 4).

Your own value is a share. 95 is refused, and so is 0.95: the most it takes is 0.9. The first message says "For 95%, type 0.95", which is then refused too (defect 5).

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Control!D21: "How far revenue must move before it counts as more or less" needs a share, such as 0.1 for 10%, from 0.01 to 0.9; got 95. For 95%, type 0.95.

Origination Cube

Extract:

[Browse...](#)

1. Set up from this extract

2. Run the cube

[Open the workbook](#)

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:
 - Control!D21: "How far revenue must move before it counts as more or less"
 needs a share, such as 0.1 for 10%, from 0.01 to 0.9; got 0.95.

Step 14: Grids

Grids shows each band crossed with each segment, one block per measure: the rate, the rate against the book, and the rate against the rest of the same band. Red is worse and green is better.

Grids									
Each grid three ways: the rate, the rate against the book, and the rate against the rest of the same band. Red is worse, green is better. For RANR more is better, so low is red.									
FICO x CHANNEL: Outcome, share of loans									
FICO (B) on loans where BAD_FLAG is 0 - as a share of all loans									
Rate	Branch	Broker	Online	All	Vs the book	Branch	Broker	Online	All
under 654	10.15%	24.47%	12.05%	12.05%	under 654	1.43x	3.43x	1.08x	1.16x
654 to under 686	8.95%	8.10%	7.72%	8.25%	654 to under 686	1.26x	1.14x	1.08x	1.16x
686 to under 712	5.44%	5.19%	4.21%	4.93%	686 to under 712	0.76x	0.73x	0.59x	0.69x
712 to under 746	4.09%	3.14%	3.90%	3.72%	712 to under 746	0.57x	0.44x	0.55x	0.52x
746 and over	4.44%	3.73%	3.68%	3.95%	746 and over	0.62x	0.52x	0.52x	0.55x
(blank)					(blank)				
(marked missing)	1.96%	12.96%	3.64%	6.25%	(marked missing)				
All	6.47%	9.04%	5.89%	7.13%	All	0.91x	1.27x	0.83x	1.00x
FICO x CHANNEL: Outcome, share of booked dollars									
FICO (B) on loans where BAD_FLAG is 0 - as a share of all booked dollars									
Rate	Branch	Broker	Online	All	Vs the book	Branch	Broker	Online	All
under 654	10.15%	24.47%	12.05%	12.05%	under 654	1.43x	3.43x	1.08x	1.16x
654 to under 686	8.95%	8.10%	7.72%	8.25%	654 to under 686	1.26x	1.14x	1.08x	1.16x
686 to under 712	5.44%	5.19%	4.21%	4.93%	686 to under 712	0.76x	0.73x	0.59x	0.69x
712 to under 746	4.09%	3.14%	3.90%	3.72%	712 to under 746	0.57x	0.44x	0.55x	0.52x
746 and over	4.44%	3.73%	3.68%	3.95%	746 and over	0.62x	0.52x	0.52x	0.55x
(blank)					(blank)				
(marked missing)	1.96%	12.96%	3.64%	6.25%	(marked missing)				
All	6.47%	9.04%	5.89%	7.13%	All	0.91x	1.27x	0.83x	1.00x
FICO x CHANNEL: GCO per booked dollar									
FICO (B) on loans where BAD_FLAG is 0 - as a share of all booked dollars									
Rate	Branch	Broker	Online	All	Vs the book	Branch	Broker	Online	All
under 654	10.15%	24.47%	12.05%	12.05%	under 654	1.43x	3.43x	1.08x	1.16x
654 to under 686	8.95%	8.10%	7.72%	8.25%	654 to under 686	1.26x	1.14x	1.08x	1.16x
686 to under 712	5.44%	5.19%	4.21%	4.93%	686 to under 712	0.76x	0.73x	0.59x	0.69x
712 to under 746	4.09%	3.14%	3.90%	3.72%	712 to under 746	0.57x	0.44x	0.55x	0.52x
746 and over	4.44%	3.73%	3.68%	3.95%	746 and over	0.62x	0.52x	0.52x	0.55x
(blank)					(blank)				
(marked missing)	1.96%	12.96%	3.64%	6.25%	(marked missing)				
All	6.47%	9.04%	5.89%	7.13%	All	0.91x	1.27x	0.83x	1.00x
FICO x CHANNEL: RANR per booked dollar (more is better)									
FICO (B) on loans where BAD_FLAG is 0 - as a share of all booked dollars									
Rate	Branch	Broker	Online	All	Vs the book	Branch	Broker	Online	All
under 654	10.15%	24.47%	12.05%	12.05%	under 654	1.43x	3.43x	1.08x	1.16x
654 to under 686	8.95%	8.10%	7.72%	8.25%	654 to under 686	1.26x	1.14x	1.08x	1.16x
686 to under 712	5.44%	5.19%	4.21%	4.93%	686 to under 712	0.76x	0.73x	0.59x	0.69x
712 to under 746	4.09%	3.14%	3.90%	3.72%	712 to under 746	0.57x	0.44x	0.55x	0.52x
746 and over	4.44%	3.73%	3.68%	3.95%	746 and over	0.62x	0.52x	0.52x	0.55x
(blank)					(blank)				
(marked missing)	1.96%	12.96%	3.64%	6.25%	(marked missing)				
All	6.47%	9.04%	5.89%	7.13%	All	0.91x	1.27x	0.83x	1.00x
FICO x CHANNEL: Outcome, share of booked dollars									
FICO (B) on loans where BAD_FLAG is 0 - as a share of all booked dollars									
Rate	Branch	Broker	Online	All	Vs the book	Branch	Broker	Online	All
under 654	10.15%	24.47%	12.05%	12.05%	under 654	1.43x	3.43x	1.08x	1.16x
654 to under 686	8.95%	8.10%	7.72%	8.25%	654 to under 686	1.26x	1.14x	1.08x	1.16x
686 to under 712	5.44%	5.19%	4.21%	4.93%	686 to under 712	0.76x	0.73x	0.59x	0.69x
712 to under 746	4.09%	3.14%	3.90%	3.72%	712 to under 746	0.57x	0.44x	0.55x	0.52x
746 and over	4.44%	3.73%	3.68%	3.95%	746 and over	0.62x	0.52x	0.52x	0.55x
(blank)					(blank)				
(marked missing)	1.96%	12.96%	3.64%	6.25%	(marked missing)				
All	6.47%	9.04%	5.89%	7.13%	All	0.91x	1.27x	0.83x	1.00x
FICO x CHANNEL: Outcome, share of booked dollars									
FICO (B) on loans where BAD_FLAG is 0 - as a share of all booked dollars									
Rate	Branch	Broker	Online	All	Vs the book	Branch	Broker	Online	All
under 654	10.15%	24.47%	12.05%	12.05%	under 654	1.43x	3.43x	1.08x	1.16x
654 to under 686	8.95%	8.10%	7.72%	8.25%	654 to under 686	1.26x	1.14x	1.08x	1.16x
686 to under 712	5.44%	5.19%	4.21%	4.93%	686 to under 712	0.76x	0.73x	0.59x	0.69x
712 to under 746	4.09%	3.14%	3.90%	3.72%	712 to under 746	0.57x	0.44x	0.55x	0.52x
746 and over	4.44%	3.73%	3.68%	3.95%	746 and over	0.62x	0.52x	0.52x	0.55x
(blank)					(blank)				
(marked missing)	1.96%	12.96%	3.64%	6.25%	(marked missing)				
All	6.47%	9.04%	5.89%	7.13%	All	0.91x	1.27x	0.83x	1.00x

Right: under 654 / Broker is 24.47%, 3.43x the book and 2.38x the rest of its band. The (marked missing) row is blank in the two comparison blocks (ringed), because its pockets have too few losses to test. The note under each block says so.

For RANR, **more is better**, and the heading says so. Low is red:

FICO x CHANNEL: RANR per booked dollar (more is better)					Vs the book					Vs the rest of its band				
Rate	Branch	Broker	Online	All	Rate	Branch	Broker	Online	All	Rate	Branch	Broker	Online	All
under 654	0.32%	0.34%	0.32%	0.33%	under 654	0.99x	1.05x	1.01x	1.02x	under 654	0.94x	1.05x	0.99x	
654 to under 686	0.31%	0.29%	0.34%	0.31%	654 to under 686	0.95x	0.95x	1.07x	0.97x	654 to under 686	0.97x	0.98x	1.16x	
686 to under 712	0.31%	0.31%	0.30%	0.30%	686 to under 712	0.95x	0.95x	0.95x	0.95x	686 to under 712	1.00x	1.00x	1.00x	
712 to under 746	0.32%	0.29%	0.35%	0.32%	712 to under 746	1.01x	0.98x	1.08x	1.00x	712 to under 746	1.01x	0.98x	1.14x	
746 and over	0.37%	0.34%	0.32%	0.35%	746 and over	1.16x	1.07x	1.00x	1.08x	746 and over	1.12x	0.99x	0.90x	
(blank)					(blank)					(blank)				
(marked missing)	0.30%	0.33%	0.23%	0.28%	(marked missing)	0.94x	1.02x	0.72x	0.88x	(marked missing)	1.09x	1.24x	0.73x	
All	0.32%	0.31%	0.33%	0.32%	All	1.01x	0.97x	1.02x	1.00x	All				

FICO x ASSET CLASS: Outcome, share of loans						Vs the book						Vs the rest of its band					
Rate	1	2	3	4	All	Rate	1	2	3	4	All	Rate	1	2	3	4	All
under 654	13.51%	11.20%	15.14%	20.41%	13.50%	under 654	1.32x	1.57x	1.13x	1.06x	1.31x	under 654	0.87x	1.00x	1.11x	1.24x	
654 to under 686	7.30%	5.92%	8.22%	11.22%	8.25%	654 to under 686	1.03x	0.83x	1.15x	1.07x	1.16x	654 to under 686	0.85x	0.86x	1.00x	1.56x	

Step 15: Materiality

Materiality shows, for every grid and measure, what each level would keep: how many pockets and how much of the grid's excess.

Materiality

What each materiality level would keep: how many pockets, and how much of the grid's excess they hold. The level itself is set on Control.

FICO x CHANNEL: Outcome, share of loans

Share of the book's loans with the outcome	Excess at or over (loans)	Pockets kept	Share of the grid's excess
0.5%	2.9	7	100%
1.0%	5.7	4	92%
2.0%	11.4	3	86%
5.0%	28.5	1	63%
10.0%	57.0	1	63%

In use: 5.7 loans

FICO x CHANNEL: Outcome, share of booked dollars

Share of the book's ORIG_BAL on loans with the	Excess at or over (ORIG_BAL dollars)	Pockets kept	Share of the grid's excess
0.5%	89,111	5	97%
1.0%	178,222	4	94%
2.0%	356,445	3	88%
5.0%	891,112	1	65%
10.0%	1,782,225	1	65%

In use: 178,222 ORIG_BAL dollars

FICO x CHANNEL: GCO per booked dollar

Share of the book's total GCO_AMT	Excess at or over (GCO_AMT dollars)	Pockets kept	Share of the grid's excess
0.5%	49,777	6	99%
1.0%	99,554	4	95%
2.0%	199,107	3	87%
5.0%	497,767	1	63%
10.0%	995,534	1	63%

In use: 99,554 GCO_AMT dollars

Right: for FICO x CHANNEL: Outcome, share of loans, the level in use is 5.7 loans (1% of the book's bad loans), which keeps 4 pockets and 92% of the excess. Control offers every level shown here.

Step 16: Check

Check records what the run did: the extract and loan count, the record file, the tie-outs, what was left out, the band edges, the loans needed to show a 1.25x gap, each materiality line with its unit, **the revenue lines and where they came from** (ringed), and every setting in words.

Check

Settings used, tie-outs, and what was left out.

Extract	Consumer book Q3.csv
Loans run	8,000
Record of this run	Consumer book Q3 - Origination Cube - what ran.yaml, beside this workbook: every setting the run used, kept for the file. It is replaced by the next Run.
Tie-out checks	132 of 132 agree: every grid adds up to the book
Left out of Outcome, share of loans	1 x BAD_FLAG flag not 0 or 1
Left out of Outcome, share of booked dollars	1 x ORIG_BAL blank; 1 x BAD_FLAG flag not 0 or 1
Left out of GCO per booked dollar	1 x GCO_AMT not a number; 1 x ORIG_BAL blank
Left out of RANR per booked dollar	1 x ORIG_BAL blank
Band edges used: FICO	654; 686; 712; 746 (5 bands)
Band edges used: ORIG_BAL	15,449; 26,324; 37,951; 49,152 (5 bands)
Band edges used: REV_DEBT	6,559; 9,837; 13,051; 17,187 (5 bands)
Loans needed for a 1.25x gap: Outcome, share of loans	about 1,756
Loans needed for a 1.25x gap: Outcome, share of booked dollars	about 2,203
Loans needed for a 1.25x gap: GCO per booked dollar	about 2,359
Loans needed for a 1.25x gap: RANR per booked dollar	about 411
Materiality line: Outcome, share of loans	5.7 loans
Materiality line: Outcome, share of booked dollars	178,222 ORIG_BAL dollars
Materiality line: GCO per booked dollar	99,554 GCO_AMT dollars
Materiality line: RANR per booked dollar	8,288 RANR_AMT dollars
Revenue counts as more or less at	1.25x and 0.80x: what luck alone can move it: in a pocket of typical size, a revenue gap smaller than 1.25x can't be told from luck at 95% sure and 80% caught
How old a loan must be to count	Every loan
Fewest loans in a pocket before it is tested	30 loans
Fewest loans with a loss before a loss rate is tested	10 losses
Smallest excess loss worth reporting	1% of the book's total losses
What a pocket is judged against	The rest of its band
How much worse than its comparison a pocket must be	1.25 times
How much better than its comparison a pocket must be	0.8 times
How far revenue must move before it counts as more or less	What luck alone can move it (suggested)
How sure a difference must be before it counts	95% sure
How often a real gap should be caught	80% of the time
Allowing for testing many pockets at once	Hold down the share of lucky finds (Benjamini-Hochberg)

Right: "132 of 132 agree", five bands for each number column, and the settings match step 9.

Step 17: Log, and Start here

Log keeps every run, newest first, including refused ones.

Log

Every Run and every refusal, newest first.

2026-09-25 18:25 Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.

2026-09-25 18:25 Couldn't run. Fix these, save, close, and press Run again:
Control!C6: "How old a loan must be to count" needs an answer. Pick one, or enter your own in column D.
Control!C13: "Fewest loans in a pocket before it is tested" needs an answer. Pick one, or enter your own in column D.
Control!C14: "Fewest loans with a loss before a loss rate is tested" needs an answer. Pick one, or enter your own in column D.

Control!C16: "Smallest excess loss worth reporting" needs an answer. Pick one, or enter your own in column D.
Control!C18: "What a pocket is judged against" needs an answer. Pick one from the list.
Control!C19: "How much worse than its comparison a pocket must be" needs an answer. Pick one, or enter your own in column D.
Control!C20: "How much better than its comparison a pocket must be" needs an answer. Pick one, or enter your own in column D.
Control!C21: "How far revenue must move before it counts as more or less" needs an answer. Pick one, or enter your own in column D.
Control!C23: "How sure a difference must be before it counts" needs an answer. Pick one, or enter your own in column D.

Columns!C3: set Checked every column to Yes once you've checked each column's meaning.

Start here now shows 0 calls left and the last run:

Origination Cube

Last run on 8,000 loans from Consumer book Q3.csv.

Step	Where	What to do
1	Control	Fill in the shaded cells: materiality, minimum loans, how much worse counts.
2	Columns	Check each column's meaning and whether to cut by it. Set C3 to Yes when done.
3	Odd values	Answer real or missing. Unanswered ones are used as is.
4	Launcher	Save and close this workbook, then press Run the cube.

Where things stand

Calls still to make on Control	0
Things to look at first on Columns	7
Odd values found	2
Last run	2026-09-25 18:25: Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.

It still says 7 things to look at on Columns, although C3 is Yes. The count only changes at the next Set up (defect 9).

Step 18: Split every pocket by a number (revolving debt)

On **Columns**, set **Split pockets by it?** to **Yes** on one number column. The walk used **REV_DEBT**. Save, close, and Run.

Columns

Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.

Checked every column? Yes

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (use ; between)	Show per pocket	Split pockets by it?	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	Loan number (key)						name contains 'loan'; a different value on every row	0%	L0000000, L0000001, L0000002
FICO	FICO score	Yes				160 rows look odd (repeated value). Used as it is until you answer it on the Odd values tab.	name contains 'fico'; 98% of values between 300 and 850	0%	-9999, 713, 716
CHANNEL	Category	Yes				suggested as Category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.	3 different values	0%	Online, Broker, Branch
ORIG_BAL BAD_FLAG	Booked amount Outcome (yes/no)	Yes				1 value is neither 0 nor 1. They're left out of the outcome rates and counted. If they mean something, fix the extract.	name contains 'bal'; positive on every loan name contains 'bad'; 0 and 1 (7.1% are 1)	0% 0%	7655.75, 27999.47, 39684.43 0, 2, 1
GCO_AMT	GCO dollars					1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.	name contains 'gco'; zero on 99% of loans, never negative	0%	0.0, #N/A, 27620.62
RANR_AMT	RANR dollars					2,612 rows look odd (negatives). Used as it is until you answer it on the Odd values tab.	name contains 'ranr'; numbers	0%	-165.2, 54.71, 38.01
ASSET_CLASS	Category	Yes				suggested as Category from its shape alone (4 different values). If it's something more specific, say so and it will be remembered.	4 different values	0%	1, 2, 4
REV_DEBT	Amount or number	Yes			Yes	suggested as Amount or number from its shape alone (numbers with 7,889 values). If it's something more specific, say so and it will be remembered.	numbers with 7,889 values	0%	15868.86, 12884.7, 8920.97

Columns

Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.

Checked every column? Yes

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (use ; between)	Show per pocket	Split pockets by it?	Look first
LOAN_NBR	Loan number (key)					
FICO	FICO score	Yes				160 rows look odd (repeated value). Used as it is until you answer it on the Odd values tab.
CHANNEL	Category	Yes				suggested as Category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.
ORIG_BAL BAD_FLAG	Booked amount Outcome (yes/no)	Yes				1 value is neither 0 nor 1. They're left out of the outcome rates and counted. If they mean something, fix the extract.
GCO_AMT	GCO dollars					1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.
RANR_AMT	RANR dollars					2,612 rows look odd (negatives). Used as it is until you answer it on the Odd values tab.
ASSET_CLASS	Category	Yes				suggested as Category from its shape alone (4 different values). If it's something more specific, say so and it will be remembered.
REV_DEBT	Amount or number	Yes			Yes	suggested as Amount or number from its shape alone (numbers with 7,889 values). If it's something more specific, say so and it will be remembered.

Origination Cube

Extract: -4db589c3cc31/scratchpad/w4/walk/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Ran on 8,000 loans from Consumer book Q3.csv; 581 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.
Split by REV_DEBT: each pocket halved at its own median. See the Split and Three-way tabs. REV_DEBT isn't cut on its own while it splits.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

Right: "581 tie-out checks agree" and "Split by REV_DEBT: each pocket halved at its own median. See the Split and Three-way tabs." The Worst for RANR line is gone, because REV_DEBT is no longer cut into bands.

Open the **Split** tab. Each grid gets a paragraph and two heat maps: the high half against the low half, and how often luck alone gives a gap that big.

Split

Every pocket split in two at its own median REV_DEBT: the high half against the low half, holding the band and segment fixed. Above 1.00x means the high half does worse (for RANR, earns more). Grids that hold fixed what REV_DEBT moves with come first. REV_DEBT isn't cut into bands of its own while it splits.

FICO x CHANNEL: Outcome, share of loans

The high-REV_DEBT half was worse in 14 of the 15 pockets big enough to test. Holding the pocket fixed, the high half has the outcome 1.95 times as often as the low half (95% range 1.72 to 2.19). Luck alone gives a gap this big under 0.01% of the time. As odds, 2.10x (1.75 to 2.53). The gap is about the same size in every pocket. This grid holds FICO fixed (REV_DEBT's correlation with it: -0.53). It doesn't hold ORIG_BAL fixed. REV_DEBT's correlation with ORIG_BAL is -0.02: 0 means unrelated, and the further from 0, the more of this gap may be ORIG_BAL.

High half vs low	Branch	Broker	Online
under 654	2.39x	2.67x	1.63x
654 to under 686	1.37x	2.92x	2.26x
686 to under 712	2.12x	1.90x	1.56x
712 to under 746	2.14x	0.78x	1.44x
746 and over	1.30x	2.33x	1.12x
(blank)			
(marked missing)			

Luck alone	Branch	Broker	Online
under 654	0.2%	under 0.01%	6.2%
654 to under 686	26.9%	0.1%	1.2%
686 to under 712	5.0%	10.5%	28.7%
712 to under 746	8.1%	61.8%	38.5%
746 and over	52.3%	6.8%	80.9%
(blank)			
(marked missing)			

FICO x CHANNEL: Outcome, share of booked dollars

The high-REV_DEBT half was worse in 15 of the 15 pockets big enough to test. Holding the pocket fixed, the high half loses 2.07x what it would at the low half's rate (95% range 1.80 to 2.34). Luck alone gives a gap this big under 0.01% of the time. This grid holds FICO fixed (REV_DEBT's correlation with it: -0.53). It doesn't hold ORIG_BAL fixed. REV_DEBT's correlation with ORIG_BAL is -0.02: 0 means unrelated, and the further from 0, the more of this gap may be ORIG_BAL.

High half vs low	Branch	Broker	Online
under 654	2.84x	2.11x	2.05x
654 to under 686	1.58x	3.82x	2.83x
686 to under 712	2.39x	2.69x	1.30x
712 to under 746	2.68x	1.12x	1.63x
746 and over	1.17x	2.82x	1.01x
(blank)			
(marked missing)			

Luck alone	Branch	Broker	Online
under 654	0.1%	under 0.01%	1.7%
654 to under 686	15.9%	0.0%	0.4%
686 to under 712	5.1%	3.2%	56.4%
712 to under 746	4.6%	84.1%	30.2%
746 and over	73.2%	5.2%	98.8%
(blank)			
(marked missing)			

FICO x CHANNEL: GCO per booked dollar

The high-REV_DEBT half was worse in 14 of the 15 pockets big enough to test. Holding the pocket fixed, the high half loses 2.06x what it would at the low half's rate (95% range 1.78 to 2.33). Luck alone gives a gap this big under 0.01% of the time. This grid holds FICO fixed (REV_DEBT's correlation with it: -0.53). It doesn't hold ORIG_BAL fixed. REV_DEBT's correlation with ORIG_BAL is -0.02: 0 means unrelated, and the further from 0, the more of this gap may be ORIG_BAL.

Split

Every pocket split in two at its own median REV_DEBT: the high half against the low half, holding the band and segment fixed. Above 1.00x means the high half does worse (for RANR, earns more). Grids that hold fixed what REV_DEBT moves with come first. REV_DEBT isn't cut into bands of its own while it splits.

FICO x CHANNEL: Outcome, share of loans

The high-REV_DEBT half was worse in 14 of the 15 pockets big enough to test. Holding the pocket fixed, the high half has the outcome 1.95 times as often as the low half (95% range 1.72 to 2.19). Luck alone gives a gap this big under 0.01% of the time. As odds, 2.10x (1.75 to 2.53). The gap is about the same size in every pocket. This grid holds FICO fixed (REV_DEBT's correlation with it: -0.53). It doesn't hold ORIG_BAL fixed. REV_DEBT's correlation with ORIG_BAL is -0.02: 0 means unrelated, and the further from 0, the more of this gap may be ORIG_BAL.

High half vs low	Branch	Broker	Online
under 654	2.39x	2.67x	1.63x
654 to under 686	1.37x	2.92x	2.26x
686 to under 712	2.12x	1.90x	1.56x
712 to under 746	2.14x	0.78x	1.44x
746 and over	1.30x	2.33x	1.12x
(blank)			
(marked missing)			

Luck alone	Branch	Broker	Online
under 654	0.2%	under 0.01%	6.2%
654 to under 686	26.9%	0.1%	1.2%
686 to under 712	5.0%	10.5%	28.7%
712 to under 746	8.1%	61.8%	38.5%
746 and over	52.3%	6.8%	80.9%
(blank)			
(marked missing)			

How to read it:

- **The first number is the one to quote.** For FICO x CHANNEL the high-debt half has the outcome **1.95 times** as often as the low half (95% range 1.72 to 2.19). The planted effect is 1.8x. The odds, 2.10x, follow.
- **Grids that hold the score fixed come first.** Each paragraph ends with what the grid holds fixed and the correlation. REV_DEBT's correlation with FICO is -0.53.
- **Blank rows** are halves under the minimum loans or losses.

Don't quote the ORIG_BAL grids. They say 2.55x and 2.57x, and the paragraph ends "It doesn't hold FICO fixed. REV_DEBT's correlation with FICO is -0.53":

ORIG_BAL x CHANNEL: Outcome, share of loans

The high-REV_DEBT half was worse in 15 of the 15 pockets big enough to test. Holding the pocket fixed, the high half has the outcome 2.55 times as often as the low half (95% range 2.26 to 2.83). Luck alone gives a gap this big under 0.01% of the time. As odds, 2.73x (2.26 to 3.29). The gap is about the same size in every pocket. This grid holds ORIG_BAL fixed (REV_DEBT's correlation with it: -0.02). It doesn't hold FICO fixed. REV_DEBT's correlation with FICO is -0.53: 0 means unrelated, and the further from 0, the more of this gap may be FICO.

High half vs low	Branch	Broker	Online	Luck alone	Branch	Broker	Online
under 15,449	1.92x	4.38x	1.90x	under 15,449	5.4% under 0.01%		8.6%
15,449 to under 26,324	1.47x	2.72x	1.63x	15,449 to under 26,324	22.8% under 0.01%		10.8%
26,324 to under 37,951	1.76x	5.85x	5.00x	26,324 to under 37,951	10.3% under 0.01%		0.0%
37,951 to under 49,152	3.67x	2.51x	2.41x	37,951 to under 49,152	under 0.01%	0.1%	1.2%

On a copy of this book with **no** revolving-debt effect, those grids still read 1.62x and 1.66x, "under 0.01%" (defect 1). The FICO grids read 1.16x and 1.15x there, with ranges that include 1.

The Three-way tab ranks every band / segment / half pocket like any other pocket, largest excess first:

Three-way

Pockets split by REV_DEBT, each tested like any other pocket, losing more than their share (RANK: earning less), largest first. The flag compares each pocket with the rest of its band. Red: worse. Amber: worse, but could be luck. "Luck alone" is how often a gap this big turns up with no real difference behind it.

Measure	Band column band		Segment column	Segment	Loans	Rate	Book rate	Excess rate	Excess is in	Material	Vs rest of book	Luck alone	Vs rest of band	Luck alone	Flag (vs the rest of its band)	Smallest gap it could show
Outcome, share of loans	FICO	under 654	CHANNEL / REV_DEBT	Broker / REV_DEBT high half	261	35.63%	7.13%	74.4 loans		yes	5.78x under 0.01%	3.27x under 0.01%				1.68x or more
Outcome, share of loans	FICO	under 654	ASSET_CLASS / REV_DEBT	4 / REV_DEBT high half	196	27.55%	7.13%	40.0 loans		yes	4.17x under 0.01%	2.66x	0.0% worse			1.80x or more
Outcome, share of loans	ORIG_BAL	15,449 to under 26,324	CHANNEL / REV_DEBT	Broker / REV_DEBT high half	210	20.48%	7.13%	28.0 loans		yes	3.03x under 0.01%	1.44x	14.2% worse, but could be luck			1.77x or more
Outcome, share of loans	FICO	under 654	ASSET_CLASS / REV_DEBT	3 / REV_DEBT high half	268	17.16%	7.13%	26.9 loans		yes	2.53x under 0.01%	1.38x	0.0% worse			1.67x or more
Outcome, share of loans	ORIG_BAL	37,951 to under 49,152	ASSET_CLASS / REV_DEBT	4 / REV_DEBT high half	175	20.00%	7.13%	22.5 loans		yes	3.92x under 0.01%	1.39x	23.7% worse, but could be luck			1.85x or more
Outcome, share of loans	FICO	under 654	CHANNEL / REV_DEBT	Branch / REV_DEBT high half	207	16.91%	7.13%	20.2 loans		yes	2.46x	0.1%	2.62x	0.1% worse		1.77x or more
Outcome, share of loans	ORIG_BAL	under 15,449	CHANNEL / REV_DEBT	Branch / REV_DEBT high half	251	14.34%	7.13%	18.1 loans		yes	2.08x	0.3%	0.94x	82.0% in line		1.70x or more
Outcome, share of loans	FICO	under 15,449	CHANNEL / REV_DEBT	Broker / REV_DEBT high half	257	13.62%	7.13%	16.7 loans		yes	1.97x	0.7%	2.54x	0.1% worse		1.69x or more
Outcome, share of loans	ORIG_BAL	26,324 to under 37,951	ASSET_CLASS / REV_DEBT	4 / REV_DEBT high half	262	13.36%	7.13%	16.3 loans		yes	1.93x	0.6%	0.87x	60.1% in line		1.68x or more
Outcome, share of loans	ORIG_BAL	26,324 to under 37,951	ASSET_CLASS / REV_DEBT	4 / REV_DEBT high half	206	15.05%	7.13%	16.3 loans		yes	2.18x	0.5%	2.87x	0.1% worse		1.78x or more

Right: FICO under 654 / Broker / high half is first: 261 loans, 35.63% bad, 5.78x the rest of the book. Rows from ORIG_BAL grids with high half in them carry the same score problem as step 18d, and this tab doesn't say so.

Step 19: Split every pocket by a category (asset class)

Clear REV_DEBT's split and set **ASSET_CLASS** instead. Save, close, and Run.

Origination Cube

Extract: -4db589c3cc31/scratchpad/w4/walk/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Ran on 8,000 loans from Consumer book Q3.csv; 382 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.
Worst for RANR per booked dollar: REV_DEBT 6,559 to under 9,837 / CHANNEL Broker.
Split by ASSET_CLASS: one layer per value. See the Three-way tab. ASSET_CLASS isn't cut on its own while it splits.
Open Consumer book Q3 - Origination Cube.xlsx: start with where it bleeds.

Right: "382 tie-out checks agree" and "Split by ASSET_CLASS: one layer per value. See the Three-way tab. ASSET_CLASS isn't cut on its own while it splits." Grids repeats each block once per asset class. **Three-way** ranks every pocket:

Three-way

Pockets split by ASSET_CLASS, each tested like any other pocket, being more than their share (RANK: earning less), targeted first. The flag compares each pocket with the rest of its band. Red: worse. Amber: worse, but could be luck. "Luck alone" is how often a gap turns up with no real difference behind it.

Measure	Band column		Segment column		Segment	Loans	Rate	Book rate	Excess	Excess is in	Material	Vs rest of book	Luck alone	Vs rest of band	Luck alone	Flag (vs the rest of its band)	Smallest gap it could show
Outcome, share of loans	FICO	under 654	CHANNEL / ASSET_CLASS	Broker / ASSET_CLASS 4	131	29.01%	7.13%	28.7 loans	yes	yes	yes	4.3x under 0.01%	2.11x	0.5% worse		4.3x or more	1.99x or more
Outcome, share of loans	REV_DEBT	17,187 and over	CHANNEL / ASSET_CLASS	Broker / ASSET_CLASS 4	124	27.42%	7.13%	25.2 loans	yes	yes	yes	4.03x under 0.01%	2.02x	1.9% worse		2.02x or more	2.09x or more
Outcome, share of loans	FICO	under 654	CHANNEL / ASSET_CLASS	Broker / ASSET_CLASS 3	111	29.73%	7.13%	25.1 loans	yes	yes	yes	4.37x under 0.01%	2.14x	0.6% worse		2.09x or more	2.09x or more
Outcome, share of loans	FICO	under 654	CHANNEL / ASSET_CLASS	Broker / ASSET_CLASS 1	145	22.07%	7.13%	21.7 loans	yes	yes	yes	3.22x under 0.01%	1.54x	16.1% worse, but could be luck		1.94x or more	1.99x or more
Outcome, share of loans	REV_DEBT	17,187 and over	CHANNEL / ASSET_CLASS	Broker / ASSET_CLASS 1	131	23.66%	7.13%	21.7 loans	yes	yes	yes	3.45x under 0.01%	1.71x	7.4% worse, but could be luck		1.99x or more	1.99x or more
Outcome, share of loans	REV_DEBT	17,187 and over	CHANNEL / ASSET_CLASS	Broker / ASSET_CLASS 3	120	24.17%	7.13%	20.4 loans	yes	yes	yes	3.52x under 0.01%	1.74x	7.4% worse, but could be luck		2.04x or more	2.04x or more
Outcome, share of loans	FICO	under 654	CHANNEL / ASSET_CLASS	Broker / ASSET_CLASS 2	136	18.38%	7.13%	15.3 loans	yes	yes	yes	2.65x 0.3%	1.25x	65.6% in line		1.97x or more	1.97x or more
Outcome, share of loans	REV_DEBT	17,187 and over	CHANNEL / ASSET_CLASS	Branch / ASSET_CLASS 4	146	15.75%	7.13%	12.6 loans	yes	yes	yes	2.26x 1.6%	1.09x	99.4% in line		1.94x or more	1.94x or more
Outcome, share of loans	FICO	under 654	CHANNEL / ASSET_CLASS	Branch / ASSET_CLASS 4	130	16.15%	7.13%	11.7 loans	yes	yes	yes	2.32x 1.7%	1.08x	98.6% in line		2.00x or more	2.00x or more
Outcome, share of loans	FICO	under 654	CHANNEL / ASSET_CLASS	Online / ASSET_CLASS 4	131	16.03%	7.13%	11.7 loans	yes	yes	yes	2.30x 1.8%	1.07x	98.0% in line		1.99x or more	1.99x or more

Right: FICO under 654 / Broker / asset class 4 first: 131 loans, 29.01% bad, flagged worse against the rest of its band. Check records the split and "Three-way pockets: 194".

Step 20: Show a median or average per pocket

On **Columns**, set **Show per pocket** to median or average on any number column. The walk put average on ORIG_BAL and median on REV_DEBT. Save, close, and Run.

FICO x CHANNEL: average ORIG_BAL per pocket*Beside the rates, for reading them: not tested, and a average doesn't add up across pockets.*

Average	Branch	Broker	Online	All
under 654	32,549	32,583	32,707	32,614
654 to under 686	32,159	32,621	31,389	32,066
686 to under 712	32,699	32,967	32,345	32,661
712 to under 746	30,751	31,887	32,569	31,747
746 and over	31,757	31,970	32,765	32,162
(blank)		27,999		27,999
(marked missing)	34,726	30,898	33,613	33,052
All	32,022	32,370	32,390	32,263

FICO x CHANNEL: median REV_DEBT per pocket*Beside the rates, for reading them: not tested, and a median doesn't add up across pockets.*

Median	Branch	Broker	Online	All
under 654	17,209	16,277	17,083	16,904
654 to under 686	13,556	13,291	13,669	13,535
686 to under 712	12,292	12,169	12,233	12,219
712 to under 746	9,952	10,431	10,659	10,272
746 and over	7,430	6,911	7,234	7,205
(blank)		12,885		12,885
(marked missing)	9,964	10,431	10,267	10,326
All	11,443	11,394	11,328	11,393

FICO x ASSET_CLASS: Outcome, share of loans*Loans where BAD_FLAG is 1, as a share of all loans.*

Rate	1	2	3	4	All
under 654	13.57%	11.20%	15.14%	20.41%	15.05%
654 to under 686	7.30%	5.92%	8.22%	11.22%	8.25%
686 to under 712	4.57%	3.04%	5.41%	6.63%	4.93%
712 to under 746	3.05%	3.48%	2.55%	5.62%	3.72%
746 and over	3.62%	4.01%	4.16%	4.02%	3.95%
(blank)		0.00%			0.00%

Right: each grid on **Grids** gets a block per column, with the note "Beside the rates, for reading them: not tested, and a median doesn't add up across pockets."

Step 21: Use the LOB's own cut points

If the LOB cuts scores at 620, 680 and 740, type them in that column's **Band edges** cell with **semicolons**: 620; 680; 740. Save, close, and Run.

Origination Cube

Extract: -4db589c3cc31/scratchpad/w4/walk/Consumer book Q3.csv

Browse...

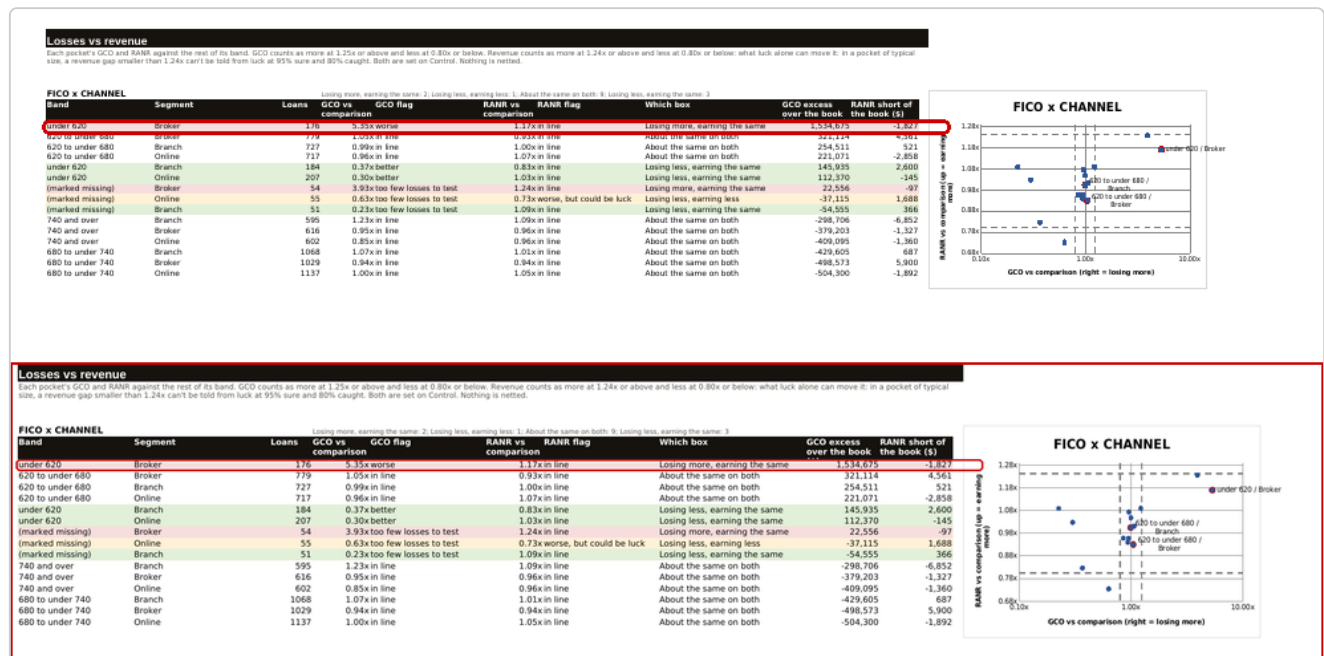
1. Set up from this extract

2. Run the cube

Open the workbook

Ran on 8,000 loans from Consumer book Q3.csv: 132 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 620 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 620 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 620 / CHANNEL Broker.
Worst for RANR per booked dollar: REV_DEBT 6,559 to under 9,837 / CHANNEL Broker.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

Right: the worst pocket is now **FICO under 620 / CHANNEL Broker**. On Losses vs revenue it reads **Losing more, earning the same**: 176 loans, GCO 5.35x the rest of its band, RANR 1.17x:



Step 22: The next extract: press Set up again

Press **1. Set up from this extract** again, on the same extract or a refreshed one with the same name.

Origination Cube

Extract: -4db589c3cc31/scratchpad/w4/walk/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Set up Consumer book Q3 - Origination Cube.xlsx from Consumer book Q3.csv:
8,000 loans, 9 columns.
2 columns to look at first, on the Columns tab.
Everything is answered. Press Run the cube.

Right: "Everything is answered. Press Run the cube." Your Control, Columns and Odd values answers are kept, and so are the last run's results and the Log:

Origination Cube

Set up from Consumer book Q3.csv: 8,000 loans, 9 columns.

Step	Where	What to do
1	Control	Fill in the shaded cells: materiality, minimum loans, how much worse counts.
2	Columns	Check each column's meaning and whether to cut by it. Set C3 to Yes when done.
3	Odd values	Answer real or missing. Unanswered ones are used as is.
4	Launcher	Save and close this workbook, then press Run the cube.

Where things stand

Calls still to make on Control	0
Things to look at first on Columns	2
Odd values found	2
Last run	2026-09-25 18:31: Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.

If the new extract has columns the old one didn't, C3 is cleared and the new columns are shaded "New since the last check" (wrong turn N).

Step 23: Prune something learned

On **Learned**, set a row's **Keep?** to **Forget**, then save, close and Run.

Learned

Meanings and answers carried over from past runs. Set a row to Forget to drop it on the next Run. To stop it being learned again, also change it on Columns.

Keep?	Kind	Column	What it learned	First confirmed	Last confirmed	Times
Keep	column	ASSET_CLASS	Category	2026-09-25	2026-09-25	10
Keep	column	BAD_FLAG	Outcome (yes/no)	2026-09-25	2026-09-25	10
Forget	column	CHANNEL	Category	2026-09-25	2026-09-25	10
Keep	column	FICO	FICO score	2026-09-25	2026-09-25	10
Keep	column	GCO_AMT	GCO dollars	2026-09-25	2026-09-25	10
Keep	column	LOAN_NBR	Loan number (key)	2026-09-25	2026-09-25	10
Keep	column	ORIG_BAL	Booked amount	2026-09-25	2026-09-25	10
Keep	column	RANR_AMT	RANR dollars	2026-09-25	2026-09-25	10
Keep	column	REV_DEBT	Amount or number	2026-09-25	2026-09-25	10
Keep	column	FICO	2000-2000-2000	2026-09-25	2026-09-25	10

Right: the window says "Forgot CHANNEL, as marked on Learned. Check it on Columns and set C3 to Yes before the next Run."

Origination Cube

Extract: -4db589c3cc31/scratchpad/w4/walk/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 620 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 620 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 620 / CHANNEL Broker.
Worst for RANR per booked dollar: REV_DEBT 6,559 to under 9,837 / CHANNEL Broker.
~~Forgot CHANNEL, as marked on Learned. Check it on Columns and set C3 to Yes before the next Run.~~
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

The next Run refuses until you do:

Origination Cube

Extract: -4db589c3cc31/scratchpad/w4/walk/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:
- Columns!C3: set Checked every column to Yes once you've checked each column's meaning.

On **Columns**, C3 is empty with "Forgotten on Learned: CHANNEL. Check what it is, then set C3 to Yes again." CHANNEL's row is shaded:

Columns									
Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.									
Checked every column? Forgotten on Learned: CHANNEL. Check what it is, then set C3 to Yes again.									
Column	What it is	Cut by it? Yes means (outcome only)	Band edges (use , between)	Show per pocket	Split pockets by it?	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	Loan number (key)						Remembered: you confirmed this as loan or application number on 2026-09-25	0%	L0000000, L0000001, L0000002
FICO	FICO score	Yes	620; 680; 740				Remembered: you confirmed this as a FICO score on 2026-09-25	0%	-9999, 713, 716
CHANNEL	Category	Yes				Forgotten on Learned: confirm what it is.	Remembered: you confirmed this as a category to cut by: channel, state, product on 2026-09-25	0%	Online, Broker, Branch
ORIG_BAL	Booked amount	Yes					Remembered: you confirmed this as booked or loan amount, which weights the reporting-level rates on 2026-09-25	0%	7655.75, 27999.47, 39684.43
BAD_FLAG	Outcome (yes/no)					1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.	Remembered: you confirmed this as the yes/no outcome: charged off, ever delinquent, decided by the system on 2026-09-25	0%	0, 2, 1
GCO_AMT	GCO dollars					1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.	Remembered: you confirmed this as gross charge-off dollars on 2026-09-25	0%	0.0, #N/A, 27620.62
RANR_AMT	RANR dollars						Remembered: you confirmed this as RANR dollars on 2026-09-25	0%	-165.2, 54.71, 38.01
ASSET_CLASS	Category	Yes					Remembered: you confirmed this as a category to cut by: channel, state, product on 2026-09-25	0%	1, 2, 4
REV_DEBT	Amount or number	Yes					Remembered: you confirmed this as a number to cut into bands on 2026-09-25	0%	15868.86, 12884.7, 8920.97

Columns									
Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.									
Checked every column? Forgotten on Learned: CHANNEL. Check what it is, then set C3 to Yes again.									
Column	What it is	Cut by it? Yes means (outcome only)	Band edges (use , between)	Show per pocket	Split pockets by it?	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	Loan number (key)						Remembered: you confirmed this as loan or application number on 2026-09-25	0%	L0000000, L0000001, L0000002
FICO	FICO score	Yes	620; 680; 740				Remembered: you confirmed this as a FICO score on 2026-09-25	0%	-9999, 713, 716
CHANNEL	Category	Yes				Forgotten on Learned: confirm what it is.	Remembered: you confirmed this as a category to cut by: channel, state, product on 2026-09-25	0%	Online, Broker, Branch

Check CHANNEL's meaning, set C3 to Yes, save, close and Run. It is learned again from your answer.

When something goes wrong

What you did	What the window says	What to do
		Clear one.

What you did	What the window says	What to do
A. Split set on two columns	"Columns!H13 and Columns!H14: only one column can split the pockets. Clear all but one."	
B. Split the only category that's cut, with the other set to Cut by it? No	"Nothing is left to cut across ... CHANNEL (Columns!H8: it splits the pockets, so it isn't a segment of its own); ASSET_CLASS (Columns!D13, Cut by it? No)."	Clear the split, or set ASSET_CLASS back to Yes.
C. average on a category	"Columns!G8: "CHANNEL" isn't a number column, so it has no average to show."	Clear the cell.
D. Edges typed as 620,680,740 and turned into one number by Excel	"Columns!F7: the band edges for "FICO" read as the single number 620,680,740 ... Type them with semicolons."	Retype with semicolons.
D2. An edge outside the column's range (620; 680; 950)	"Columns!F7: FICO runs from 496 to 921, so an edge at 950 would leave a band empty."	Type edges inside that range.
E. Workbook still open in Excel	"... is open in Excel. Close it, then press Run again." Set up says the same. Nothing else changes.	Close Excel and press again.
F. A column renamed in the extract after Set up	"Couldn't run: the extract has no column "CHANNEL" (a segment) ... If a column was renamed or dropped, press Set up again". After Set up: "New columns since the last check: CHNL. Columns!C3 needs a Yes again."	Press Set up, check the new name, set C3 to Yes, Run.
G. Rows changed in the extract after Set up	"Consumer book Q3.csv has changed since Set up...", then it runs on the new rows (6,000).	Check the loan count.
H. Workbook copied into a new folder with a new extract of the same name	The same "has changed since Set up" line; it runs the new folder's extract (3,000 loans).	Press Set up in the new folder first.
I. Split set on GCO, the loan number or the outcome	"Columns!H11: "GCO_AMT" is marked GCO dollars, which can't split the pockets. Only a score, ratio, amount or category can." The key and the outcome are refused the same way.	Split only by something known at booking.
I2. Split set on the booked amount (ORIG_BAL)	Runs, 571 tie-outs (defect 6).	Treat with care: it takes ORIG_BAL out of the bands.
J. A dollar amount typed as your own materiality (100000)	Runs. Check says "\$100,000 of GCO" and "no line" for every other rate, with a warning each.	Use it for GCO only, as it says.
K. The workbook picked as the extract	"... is the workbook, not the loan file. Pick the extract it was set up from (Consumer book Q3.csv or Consumer book Q3.xlsx)." Pressing Run then says "There's no workbook ... Press 1. Set up" (defect 11).	Browse to the .csv.
L. 95 typed for "how sure"	"... needs a share, such as 0.95 for 95%, from 0.5 to 0.999; got 95. For 95%, type 0.95."	Type 0.95, or pick "95% sure".
M. 95, or 0.95, typed as your own revenue line	"... needs a share, such as 0.1 for 10%, from 0.01 to 0.9; got 95. For 95%, type 0.95." Then 0.95 is refused too.	Type a share up to 0.9, such as 0.15.

What you did	What the window says	What to do
N. New columns in a refreshed extract	"New columns since the last check: CURR_STATUS, DTI. Columns!C3 needs a Yes again."	Check them, set C3 to Yes.
O. Forget on Learned, then Run twice	The second Run: "Columns!C3: set Checked every column to Yes..."	Check the forgotten column, set C3 to Yes.

A: two splits

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!H13 and Columns!H14: only one column can split the pockets. Clear all but one.

B: the only segment split

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Nothing is left to cut across: mark at least one column as a category (or term) on the Columns tab, with Cut by it set to Yes. Changed from what was suggested: CHANNEL (Columns!H8: it splits the pockets, so it isn't a segment of its own); ASSET_CLASS (Columns!D13, Cut by it? No).

C: average on a category

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!G8: "CHANNEL" isn't a number column, so it has no average to show. Clear the cell, or change what it is.

D and D2: edges

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:
- Columns!F7: the band edges for "FICO" read as the single number 620,680,740.
Excel dropped the commas. Type them with semicolons: 620; 680; 740.

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:
- Columns!F7: FICO runs from 496 to 921, so an edge at 950 would leave a band empty. Type edges inside that range, with semicolons: 620; 680; 740.

E: workbook open. Simulated by locking the file (chattr +i), since Excel isn't available here. Run and Set up both refuse, and the memory file doesn't change:

Origination Cube

Extract:

Consumer book Q3 - Origination Cube.xlsx is open in Excel. Close it, then press Run again.

Origination Cube

Extract:

Consumer book Q3 - Origination Cube.xlsx is open in Excel. Close it, then press Set up again.

F: a column renamed after Set up, then Set up pressed

Origination Cube

Extract:

[Browse...](#)

[1. Set up from this extract](#)

[2. Run the cube](#)

[Open the workbook](#)

Couldn't run: the extract has no column "CHANNEL" (a segment). Its columns are: LOAN_NBR, FICO, CHNL, ORIG_BAL, BAD_FLAG, GCO_AMT, RANR_AMT, ASSET_CLASS, REV_DEBT. If a column was renamed or dropped, press Set up again

Origination Cube

Extract:

[Browse...](#)

[1. Set up from this extract](#)

[2. Run the cube](#)

[Open the workbook](#)

Set up Consumer book Q3 - Origination Cube.xlsx from Consumer book Q3.csv: 8,000 loans, 9 columns.
New columns since the last check: CHNL. Columns!C3 needs a Yes again.
3 columns to look at first, on the Columns tab.
Next: fill in the shaded cells on Columns, save, close, and press Run.

G: fewer rows after Set up

Origination Cube

Extract:

Consumer book Q3.csv has changed since Set up. If columns were added or renamed, press Set up first.
Ran on 6,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.
Worst for RANR per booked dollar: FICO 714 to under 747 / ASSET_CLASS 2.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

H: the workbook copied to another folder

Origination Cube

Extract:

Consumer book Q3.csv has changed since Set up. If columns were added or renamed, press Set up first.
Ran on 3,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 653 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 653 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 653 / CHANNEL Broker.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

I: split by GCO, the loan number, the outcome; I2: by the booked amount

Origination Cube

Extract:

[Browse...](#)

[1. Set up from this extract](#)

[2. Run the cube](#)

[Open the workbook](#)

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!H11: "GCO_AMT" is marked GCO dollars, which can't split the pockets. Only a score, ratio, amount or category can.

Origination Cube

Extract:

[Browse...](#)

[1. Set up from this extract](#)

[2. Run the cube](#)

[Open the workbook](#)

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!H6: "LOAN_NBR" is marked Loan number (key), which can't split the pockets. Only a score, ratio, amount or category can.

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:
- Columns!H10: "BAD_FLAG" is marked Outcome (yes/no), which can't split the pockets. Only a score, ratio, amount or category can.

Origination Cube

Extract:

Ran on 8,000 loans from Consumer book Q3.csv; 571 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.
Worst for RANR per booked dollar: REV_DEBT 6,559 to under 9,837 / CHANNEL Broker.
Split by ORIG_BAL: each pocket halved at its own median. See the Split and Three-way tabs. ORIG_BAL isn't cut on its own while it splits.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

J: your own dollar materiality

Origination Cube

Extract:

Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.
Worst for RANR per booked dollar: REV_DEBT 6,559 to under 9,837 / CHANNEL Broker.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

Check

Settings used, tie-outs, and what was left out.

Extract	Consumer book Q3.csv
Loans run	8,000
Record of this run	Consumer book Q3 - Origination Cube - what ran.yaml, beside this workbook: every setting the run used, kept for the file. It is replaced by the next Run.
Tie-out checks	132 of 132 agree: every grid adds up to the book
Left out of Outcome, share of loans	1 x BAD_FLAG flag not 0 or 1
Left out of Outcome, share of booked dollars	1 x ORIG_BAL blank; 1 x BAD_FLAG flag not 0 or 1
Left out of GCO per booked dollar	1 x GCO_AMT not a number; 1 x ORIG_BAL blank
Left out of RANR per booked dollar	1 x ORIG_BAL blank
Band edges used: FICO	654; 686; 712; 746 (5 bands)
Band edges used: ORIG_BAL	15,449; 26,324; 37,951; 49,152 (5 bands)
Band edges used: REV_DEBT	6,559; 9,837; 13,051; 17,187 (5 bands)
Loans needed for a 1.25x gap: Outcome, share of loans	about 1,756
Loans needed for a 1.25x gap: Outcome, share of booked dollars	about 2,203
Loans needed for a 1.25x gap: GCO per booked dollar	about 2,359
Loans needed for a 1.25x gap: RANR per booked dollar	about 411
Materiality line: Outcome, share of loans	no line: the dollar line on Control is a GCO amount
Materiality line: Outcome, share of booked dollars	no line: the dollar line on Control is a GCO amount
Materiality line: GCO per booked dollar	100,000 GCO_AMT dollars
Materiality line: RANR per booked dollar	no line: the dollar line on Control is a GCO amount
Revenue counts as more or less at	1.25x and 0.80x: what luck alone can move it: in a pocket of typical size, a revenue gap smaller than 1.25x can't be told from luck at 95% sure and 80% caught
How old a loan must be to count	Every loan
Fewest loans in a pocket before it is tested	30 loans
Fewest loans with a loss before a loss rate is tested	10 losses
Smallest excess loss worth reporting	\$100,000 of GCO
What a pocket is judged against	The rest of its band
How much worse than its comparison a pocket must be	1.25 times
How much better than its comparison a pocket must be	0.8 times
How far revenue must move before it counts as more or less	What luck alone can move it (suggested)
How sure a difference must be before it counts	95% sure
How often a real gap should be caught	80% of the time
Allowing for testing many pockets at once	Hold down the share of lucky finds (Benjamini-Hochberg)
Warning	Outcome, share of loans: no materiality line. The dollar line is a GCO amount, so it is only applied to GCO
Warning	Outcome, share of booked dollars: no materiality line. The dollar line is a GCO amount, so it is only applied to GCO
Warning	RANR per booked dollar: no materiality line. The dollar line is a GCO amount, so it is only applied to GCO

K: the workbook picked as the extract, then Run

Origination Cube

Extract:

Consumer book Q3 - Origination Cube.xlsx is the workbook, not the loan file. Pick the extract it was set up from (Consumer book Q3.csv or Consumer book Q3.xlsx).

Origination Cube

Extract:

There's no workbook for Consumer book Q3 - Origination Cube.xlsx yet. Press 1. Set up from this extract first.

L: 95 for "how sure"

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:
- Control!D23: "How sure a difference must be before it counts" needs a share, such as 0.95 for 95%, from 0.5 to 0.999; got 95. For 95%, type 0.95.

N: new columns in a refreshed extract

Origination Cube

Extract:

Set up Consumer book Q3 - Origination Cube.xlsx from Consumer book Q3.csv:
8,000 loans, 11 columns.
New columns since the last check: CURR_STATUS, DTI. Columns!C3 needs a Yes again.
4 columns to look at first, on the Columns tab.
Next: fill in the shaded cells on Columns, save, close, and press Run.

Columns

Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under look first. Set C3 to Yes when done, confirmed meanings carry over to the next extract.

Checked every column? New since the last check: CURR_STATUS, DTI. Check them, then set C3 to Yes again.

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (use ; between)	Show per pocket	Split pockets by it?	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	Loan number (key)						Remembered: you confirmed this as loan or application number on 2026-09-25	0%	L0000000, L0000001, L0000002
FICO	FICO score	Yes					Remembered: you confirmed this as a FICO score on 2026-09-25	0%	-9999, 713, 716
CHANNEL	Category	Yes					Remembered: you confirmed this as a category to cut by: channel, state, product on 2026-09-25	0%	Online, Broker, Branch
ORIG_BAL	Booked amount	Yes					Remembered: you confirmed this as booked or loan amount, which weights the reporting-level rates on 2026-09-25	0%	7655.75, 27999.47, 39684.43
BAD_FLAG	Outcome (yes/no)					1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.	Remembered: you confirmed this as the yes/no outcome: charged off, ever delinquent, decided by the system on 2026-09-25	0%	0, 2, 1
GCO_AMT	GCO dollars					1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.	Remembered: you confirmed this as gross charge-off dollars on 2026-09-25	0%	0.0, #N/A, 27620.62
RANR_AMT	RANR dollars						Remembered: you confirmed this as RANR dollars on 2026-09-25	0%	-165.2, 54.71, 38.01
ASSET_CLASS	Category	Yes					Remembered: you confirmed this as a category to cut by: channel, state, product on 2026-09-25	0%	1, 2, 4
REV_DEBT	Amount or number	Yes					Remembered: you confirmed this as a number to cut into bands on 2026-09-25	0%	15868.86, 12884.7, 8920.97
CURR_STATUS	Servicing data					New since the last check.	name contains 'status'	0%	C, D36, CD
DTI	DTI style rate	Yes				New since the last check.	name contains 'dti'; a ratio with median 40%	0%	0.20, 0.24, 0.22

How this was walked

On a Linux test machine:

- **The window:** Origination Cube.pyw's window on a virtual display under Python 3.12, its buttons pressed by driver/drive.py (through driver/win.sh), each state photographed.
- **Excel's part:** done cell by cell with openpyxl (driver/edit.py), standing in for picking from the dropdowns and typing.
- **The workbook:** each state printed through LibreOffice (driver/render.sh) and photographed page by page. driver/shots.py crops and rings every picture, finding each ringed row by its own words on the printed page.
- **Wrong turns and revenue options:** each in a folder of its own, from a copy of the answered workbook (driver/wrong.sh).
- **The build:** frozen at commit 1156ff4 with git archive.
- **Memory and preferences:** a scratch CUBE_MEMORY and HOME.
- **The no-effect check** behind step 18's warning: driver/noplant.py, and the same book run through the window (wrong-noplant-split.png).
- **Reading tabs back:** driver/boxcheck.py (Losses vs revenue boxes against flags and dollars), driver/revenue_options.py (step 13's table), driver/threeway.py (Three-way flags by grid), driver/luckline.py (what the suggested revenue line is made of).

The order walked: steps 1 to 17, the revenue options, then 18 to 21 on the same workbook, then 22 and 23, then the wrong turns.